



Recent progress in marine foul-release polymeric nanocomposite coatings



M.S. Selim ^{a,b,1}, M.A. Shenashen ^{a,b,1}, Sherif A. El-Safty ^{a,c,*}, S.A. Higazy ^b, M.M. Selim ^d, H. Isago ^a, A. Elmarakbi ^e

^a National Institute for Materials Science (NIMS), 1-2-1 Sengen, Tsukubashi, Ibaraki-ken 305-0047, Japan

^b Petroleum Application Department, Egyptian Petroleum Research Institute (EPRI), Nasr City 11727, Cairo, Egypt

^c Graduate School for Advanced Science and Engineering, Waseda University, 3-4-1 Okubo, Shinjuku-ku, Tokyo 169-8555, Japan

^d Department of Mathematics, Al-Aflaj College of Science and Human Studies, Prince Sattam Bin Abdulaziz University, Al-Aflaj 710-11912, Saudi Arabia

^e Automotive Composites Group, Faculty of Engineering and Advanced Manufacturing, University of Sunderland, Sunderland SR6 0DD, UK

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ABSTRACT

Progress in materials science is associated with the development of nanomaterials in terms of energy-saving, environmentally friendly, and low-cost methods. Since the use of tributyltin compounds in antifouling coatings was banned in 2003, the search for ecofriendly alternatives has been promoted. Foul-release (FR) nanocoatings have been extensively investigated because of their non-stick, ecological, and economic advantages. Such nanocomposite systems are dynamic non-stick surfaces that deter any fouling attachment through physical anti-adhesion terminology. Instead of biocidal solutions, several functional FR nanocomposite coatings have been developed to counter biofouling and biocorrosion with ecological and ecofriendly effects. Selected inorganic nanofillers have been incorporated because of their enhanced interaction at the filler-polymer interface for nanocomposites. Metallic nanoparticles and their oxides have also been widely explored because of their unique morphological characteristics and size-dependent, self-cleaning properties. In modeling a novel series of FR nanocoatings, two modes of prevention are combined: chemical inertness and physical microfouling repulsion for maritime navigation applications. Long-term durability and self-cleaning performance are among the advantages of developing effective, stable, and ecofriendly modeling alternatives. This review provides a holistic overview of nano-FR research achievements and describes recent advancements in non-stick marine nanocoatings for ship hulls. This review highlights the key issues of nanocomposite structures and their features in improving the biological activity and surface self-cleaning performance of ship hulls. This review may also open new horizons toward futuristic developments in FR nanocomposites for maritime navigations.

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* Corresponding author at: Research Center for Strategic Materials, National Institute for Materials Science (NIMS), 1-2-1 Sengen, Tsukubashi, Ibaraki-ken 305-0047, Japan.

E-mail addresses: sherif.elsafty@nims.go.jp, sherif@aoni.waseda.jp (S.A. El-Safty).

URLs: <http://www.nims.go.jp/waseda/en/labo.html>, <http://www.nano.waseda.ac.jp> (S.A. El-Safty).

¹ Both authors are equally contributed in the production of this work.

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1. Introduction

Nanocomposites constitute a class of materials that exhibit advanced properties at low nanoparticle (NP) concentrations in comparison with conventional filler contents in coatings [1]. Selected NPs in coating formulations can enhance various properties, including antifouling (AF), mechanical and optical characteristics, permeability, and wettability [2]. Biofouling is a complex issue that causes serious economic casualties and unfavorable ecological impacts on maritime ecosystems. Marine fouling costs about US\$ 150 billion every year in transportation [3]. As such, fouling control through ecofriendly and economic AF paint solutions is of great importance because the annual consumption reaches 80,000 tons [3]. Fouling layers increase friction drag and hydrodynamic weight and thus decrease shipping speed and maneuverability [4]. For this reason, high fuel consumption is necessary to maintain the required speed and navigation setting; as a consequence, financial costs are increased and harmful compounds are emitted into the environment. This action also requires personnel resources, entails machinery effort, consumes time, and generates wastes that pose harmful effects on health and environment [5]. Therefore, these problems have prompted researchers to develop coatings that prevent ship hull biofouling caused by commercial biocidal AF coatings [6,7]. With environmental toxicity issues related to the use of toxic AF coatings, modern research has focused on environmentally friendly alternatives, such as foul-release (FR) technology [8]. This review discusses the advanced ecofriendly technologies for AF coatings and focuses on the modern streams achieved in FR coatings for maritime applications.

Non-stick FR coatings, which include fluoropolymers and silicon compounds, offer an ecofriendly alternative that can be used for prolonged periods [9]. FR coating technology acts by preventing fouling settlements and providing extremely smooth self-cleaning surfaces [10]. Organo-silicone polymers, particularly polydimethylsiloxane (PDMS), are more efficient than fluoropolymers and thus are considered the most promising FR coating systems [4–6,8,11,12]. As a non-stick FR layer, PDMS with a Si–O backbone and a CH₃ side chain possesses excellent properties, such as high level of smoothness, hydrophobicity, and mobile molecular structure; however, surface tension and porosity are low [12,13]. In addition, PDMS exhibits excellent heat resistance, anti-oxidation and anti-ozone properties, and durability against ultraviolet (UV) irradiation [14]. Although PDMS coatings show inherently superior FR attributes, its combination with inorganic nano-additives has become necessary. An enhanced matrix-nanofiller interaction provides a cost-effective method and improves some characteristics of nanomaterials. Several characteristics can influence the nanocomposite FR features, such as filler type, size, shape, dispersion percentage, and compatibility with matrix segments [15–17]. Reduced surface tension and increased contact angle (CA) are also major factors that diminish fouling settlements and bacterial cohesion [18]. Coating techniques with noble metal NPs and metal oxides are possibly effective methods to prevent fouling because of their stability against microbial attacks [18–20]. Among these materials, Cu₂O, ZnO, TiO₂, and Ag NPs are easily prepared from low-cost, highly secured natural sources and thus are applicable to feasibility and advanced studies [20]. A series of uniform in situ, ex situ, and sol-gel FR polymer nanocomposites with varying degrees of hydrophobicity and hydrophilicity has been investigated in micro- and macro-biofouling assays and in field tests [5,21]. In addition to surface chemistry, surface free energy and surface topologies affect adhesion mechanism, biofouling, and self-cleaning performance [22]. Considering the high costs associated with fouling

problems, toxic AF paints, and mitigation programs worldwide, we understand that the proper utilization of modeled non-toxic FR nanocoatings may provide an environmentally friendly alternative. In the following sections, the main achievements in FR nano-coatings are summarized and arguments are provided to support our conclusion that this nano-coating technology is a well-established platform for the development of ecofriendly coatings that hinder fouling in maritime activities. In this context, in-depth research programs for the design of green and cost-effective fouling preventive measures have been implemented. Alternative solutions have also been investigated. FR coatings composed of nanostructured hydrophilic/hydrophobic surfaces have also been widely explored.

2. Biofouling

Marine biofouling refers to the undesirable colonization of man-made structures immersed in seawater by biotic and abiotic dissolved compounds, microorganisms, plants, and animals [3,12,23]. Fouling can be considered a noise to various economic and environmental issues [24]. Numerous fouling organisms may be divided into two groups on the basis of their size: (1) microorganisms, also known as biofilm, slime, or microfouling and (2) macrofouling (Fig. 1). Fouling organisms undergo five main stages of succession, namely, adsorption, immobilization, consolidation, microfouling, and macrofouling [25]. A biofilm can be described as a collection of microbial strains embedded in a natural polymer matrix secreted by microorganisms referred to as extracellular polymeric substances [26]. Bacteria that form slime films are present after the attached bacterial cells reproduce by dividing into daughter cells. Each of the resulting daughter cells then grows to normal size and divides again [27]. Microbial strains must form biofilms, which protect them from natural stress factors, such as pH shift, UV radiation, osmotic shock, and desiccation and provide a system with high immunity against biocidal and antibacterial attack [28]. Biofilms are controlled by physical forces to form a complex structure composed of cells, aggregates, and bacterial microcolonies [29,30]. Moreover, they are randomly located in a highly hydrated polymeric matrix (80–95% water) with pores or voids, through which nutrient flow is measured. Biofilms are further characterized by their heterogeneity in coverage and thickness, and their sizes range from several micrometers to several centimeters. Fouling is inevitable for almost any material immersed in seawater. Fouling development can be divided into three steps [31].

Molecular fouling is the primary cause of the adsorption of organic and inorganic macromolecules, such as polysaccharides and proteins from the marine biosphere. Microfouling involves two steps, namely, primary and secondary colonization [32]. In primary colonization, primary colonizers, including bacteria and diatoms, adsorb onto a conditioning layer via physical forces; this adsorption is reversible because bacteria can be easily removed [33]. Numerous primary colonizers can create extracellular polysaccharides and irreversibly attach to a conditioning film through adhesion. In secondary colonization, secondary colonizers consist of macroalgal spores and protozoa, which are considered parts of a microfilm. An interesting feature of microfouling is that the microfilm mass progresses nonlinearly. Macrofouling is of greater concern than microfouling because of the larger contribution to weight or hydrodynamic loading of the former than the latter. Macrofouling may also be caused by plants or animals and can be classified into hard and soft fouling. As such, ship hull fouling has been extensively explored [34].

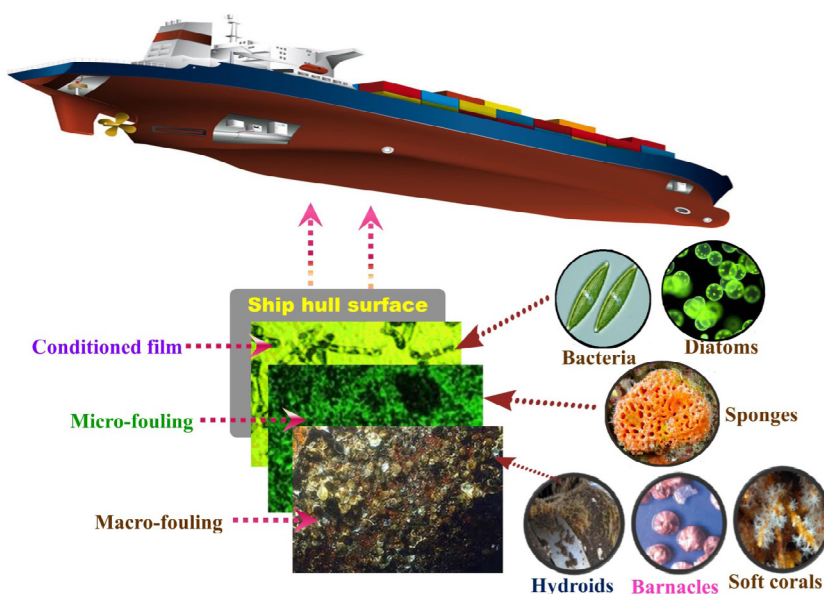


Fig. 1. A schematic illustration of the three common fouling steps including molecular (conditioned film), micro- and macro-fouling.

3. Fouling-related costs

Marine biofouling causes environmental disasters and consumes billions of dollars every year in the shipping industry (Fig. 2) [35]. Shipping is a prime facilitator of global trade and economic growth because it represents more than 90% of the international commerce [36]. Surface smoothness and drag are decreased when organisms attach to vessel hulls; as a consequence, hydrodynamic weight is increased, top speed is subsequently reduced, and maneuverability is lost [37]. Therefore, fouling increases fuel consumption by up to 40% and promotes the emission of harmful compounds, such as SO_x and NO_x , which cause acid rains and soil damage; other toxic substances, such as CO_2 and atmospheric pollutants, are also released and pose serious environmental problems [38]. Approximately 60,000 people lose their lives and €200 billion is lost every year because of the harmful emissions from global maritime navigation [39]. Such an increase in fuel consumption triggered by fouling can increase the total cruise expenses by approximately 77% [40] and can shorten the dry-docking intervals.

AF coating technologies have been developed because of adverse ecological and economic impacts of biofouling and defects in nanocoating techniques. Fouling settlements are conventionally inhibited by toxicants released from the applied biocidal AF paints, but their toxicity severely affects non-objective organisms such as dolphins [41]. This market is worth billions of dollars annually worldwide, but the assessment of total costs is difficult as this problem involves numerous industrial factors; the poor efficiency of AF efforts also requires frequent and continuous counter measures [42]. Fouling costs are related to several major effects, including physical material damage, mechanical interference, biological competition for resources, such as food and space, environmental modification caused by the colonization of culture infrastructure, and increased friction and drag resistance on equipment. Equipment maintenance and loss directly contribute to production costs; in particular, biofouling increases capital cost and accounts for approximately 20% of overall fouling in energy generation [43]. In addition to interferences in structural functions, the corrosion of metallic surfaces may be accelerated or paint coatings protecting the surfaces from rusting may become damaged. Furthermore, the treatment of wastewater contaminated with biocidal AF additives is an emerging cost factor because the amount of released biocides increases; as such, these additives should be removed [44].

Full-scale ship tests have been conducted to determine the effect of fouling on the drag of AF coatings and have revealed that even slime can significantly increase drag to a dangerous extent. Therefore, technologies should be improved to protect ships from fouling, to extend the period between dockings, and to save time and costs [45]. However, statistical data on the volumes and value of AF paints have yet to be provided. According to a report from industrial representatives who provided at least 56 data validated by European Chemicals Agency in 2009, the global market consumed 904 million litres of AF paints costing approximately €3.5 billion in 2012 [6,46]. Market analysis has forecasted that the global market of marine coatings will increase twofold by 2018 and exhibit an annual growth rate of >11% from 2013 to 2018 despite the ongoing severe crisis of global shipping; for example, German ship owners, who are the largest stakeholders in the worldwide container shipping trade with 1600 container vessels, reduced the national merchant fleet by 200 vessels between 2012 and 2013 [6,47]. This phenomenon also occurred because ship owners focused on cost reduction to address the historically high costs of oil-derived fuel. Another issue involves the economic sustainability of elastomeric coatings. When these coatings are applied to large vessels that consume tens of thousands of US dollars of fuel per day, the large return on investments associated with energy savings largely justifies the high capital expense to apply new silicone-derived coatings on dry dock. For instance, ePaint, which is a small US company, manufactures 10,000 gallons per year of ZnO-based (5% as an active biocide) photocatalytic AF formulated paints and SeaNine 211 to repel hard-shelled organisms through the formation of a thin H_2O_2 layer around the hull [6,48].



Fig. 2. Fouling costs upon the attachment to ship hull which cause serious problems in shipping industry.

Table 1

Historical development of the AF strategies [35,43].

Timeline	Major events
700 BCE	Phoenicians and Carthaginians: Pitch, lead sheathing
400 BCE	Arsenic, sulfur in oil
200 BCE	Greeks, Romans: Lead sheathing and copper nails
1000 CE	Vikings: seal tar
1200–1500 CE	Pitch, oil, resin, tallow (Columbus's ship)
1500 CE	Wooden sheathing
1500–1700 CE	Lead sheathing
1618 CE	Underwater Copper: Christian IV
1625 CE	1st Coating patent by William Beale [49]
1758 CE	Copper sheathing (HMS Alarm)
<i>Iron ships</i>	
1800–1900s A.D.	Heavy metals (copper, arsenic, mercury) incorporated into coatings
1860 CE	Hot plastic paints
20th century CE	Cold plastic paints
1950s A.D.	Development of tri-organotin (R) ₃ -Sn conventional coatings
1976 CE	TBT-SPC technology, 1976 patent, allowed control of biocide release rates [43]
1977 CE	First FR coating patent [43c]
1980s A.D.	– Development of TBT-SPC coatings allowed control of biocide release rates – TBT linked to shell abnormalities in oysters and imposex in dog whelks
1987–1990 CE	TBT coatings prohibited on vessels in some countries
1990s A.D.	Copper release rate restrictions introduced in Denmark and considered elsewhere e.g. California, USA
2000s A.D.	Research into environmentally friendly AF alternatives increases
2001 CE	International Maritime Organization (IMO) adopts "AFS Convention" to eliminate TBT from AF coatings from vessels through: 2003 – prohibition of further application of TBT 2008 – prohibition of active TBT presence
2001 CE – present	FR coating technology and nanocomposites

4. History of fouling prevention

The development of fouling prevention solutions is closely associated with increased marine transportation and shipping of people and cargo. The harmful effects of fouling on ship performance have been documented, and attempts to utilize AF systems began in the 7th century BCE (Table 1). Early written testimonies on the first AF attempts were provided by Phoenicians, Carthaginians, Greeks, and Romans, which were ancient Mediterranean civilizations [49,50]. A lead-sheathed timber was first used in Phoenician galleys in 7th century BCE. In 5th century BCE, arsenic and sulfur-and-oil mixture coatings were utilized to prevent fouling [50]. In the 3rd century, lead sheathing was secreted with copper nails by Romans and Greeks [50]. Several centuries later, the Vikings reportedly used "seal tar." In the 13–15th centuries, pitch was widely utilized to prevent fouling and usually mixed with oils or tallow as used by Christopher Columbus during his expedition to the New World [51]. Hide sheathing was utilized in the 14th century. In the 16th century, ships were sheathed with woods placed above a coated film of animals' tar [51]. Although lead sheathings were the most commonly used among AF systems, their use resulted in severe rusting of steel materials, such as ship rudders; thus, the use of lead sheathings was banned by the British in the 17th century [43].

Wooden sheathing was used again, coated with many blends, such as iron and copper spike fillers in tar or grease. This approach was followed by copper sheathing, which was an important stage in the development of AF. In the first patent published in 1625 by William Beale, a British man, who based his formula on cement admixture including iron and copper compositions, copper compounds were used as AF agents [49]. In 1758, AF sheathing with copper was investigated in England and yielded more efficient results than lead sheathing did; therefore, the technology had been extensively utilized by the governmental fleet since 1758 and soon became the preferred approach to protect wooden ships hulls [49,51]. However, the advent of steel ships and the subsequent corrosion problems associated with metallic sheathing have prompted ship owners to stop the application of this technology and focused on the use of AF compositions. In the mid-1800s, William John Hay isolated iron hull from metallic toxicants by using a non-conductive varnish and copper oxide, mercury oxide, and arsenic dispersed in high amounts of resins [35]. Moreover, Mallet recorded a patent of AF coating, a blend of varnish and toxicants with high solubility, but the developed system suffered from abrasion and uncontrollable solution rate [35]. In 1860, hot-plastic paint was developed by using copper compounds embedded in rosin admixture as a second layer after an anticorrosive layer was applied. Nevertheless, these coatings entailed high cost, yielded a comparatively inefficient performance, and provided a short life span [49].

Paint for the bottom part of a ship was first manufactured by the Americans in 1908 after a spirit varnish paint was successfully developed; many paint developments were based on mercury oxide dispersed in graded grain alcohol and turpentine; as a result, the lifetime of the manufactured paint was approximately 9 months [52]. According to a research study conducted by the US Navy in the early 20th century, rosin demonstrated advantageous properties, such as low cost,

abundant source, and successful replacement for the rare and costly shellac. Hence, rosin was blended with poisonous compositions of mercury or copper for enhanced performance; for instance, rosin was combined with synthetic petroleum-derived resins, which enhanced their mechanical characteristics but caused concerns on safety and health [43]. Hot-plastic paints are difficult to apply; thus, cold-plastic paints were developed for convenient application in ship sites. These paints resisted fouling, and the period between dry dockings was doubled.

In the mid-20th century, the exacerbated poisoning of organotin compounds, especially tributyltin (TBT), was successfully recorded by van der ker's team, which presented excellent AF properties of the TBT moiety. Kiil et al. [53] modeled the working mechanisms of these paints; in this study, a TBT biocidal copolymer hydrolyzes and erodes the toxicants in the marine environment. TBT compounds were initially utilized in advanced copper AF coatings and progressively applied as a whole AF system, which underwent "free association form" inside the matrix. TBT self-polishing coatings (TBT-SPC), which are composed of TBT acrylate esters as AF coatings, were developed in 1958. James patented a SPC based on copolymers of TBT acrylate and methyl methacrylate in 1964 [43,54]. In general, TBT-SPC technology addressed the rapid leaching of TBT-bearing compounds incorporated in free-associated form; thus, long-lasting AF protection was provided (Fig. 3) [55]. TBT-SPC provides various advantages, such as smooth hydrodynamic profile during sailing, decreased fuel consumption, and constant biocide release rates over time. The hulls of more than 70% of the worldwide fleet of ships were painted with TBT-SPCs because of the success of this technology in the 1990s. Unfortunately, the degradation rate of TBT-bearing biocides was slow in the sea water column; as a consequence, lipophilic compounds have become readily bioavailable to non-target species [52,54]. Once these compounds are partitioned into cellular membranes, they disrupt essential metabolic and enzymatic functions, such as energy production through oxidative phosphorylation; as a result, organisms die [56]. At considerably low concentrations, these compounds cause a series of sublethal effects, such as endocrine disruption leading to sexual disorders, including imposex [56,57]. Moreover, the International Maritime Organization (IMO) reported that these compounds accumulate in mammals and debilitate immunological defenses in fishes [54]. The IMO prohibited the application of organo-tin-based AF coatings on all vessels on January 1, 2003 (Fig. 4) [55,58]. Copper-based tin-free AF paints were developed, but the use of these paints has been under scrutiny in several countries and has been subjected to environmental restrictions in terms of copper release rates. Despite the high toxicity of copper to numerous maritime communities, common algae exhibit a high degree of tolerance and can easily attach to the surface. Thus, booster biocides were added to the copper AF coatings to overcome these problems and afford highly biocidal coatings, such as Irgarol 1051 and diuron [58,59]. These paints often contain herbicides, which slow down the growth rate of photosynthetic organisms. As such, legislation in certain countries has been implemented to regulate the application of these booster toxicants because of their exacerbated poisoning effect even at small percentages and non-biodegradability in water [60]. The erosion of AF vessel coatings causes surface damages, such as rusting and discoloration, complete leaching of ingredients, and short dry-docking intervals. With the growing concern on the harmful side effects of biocidal coatings on the environment, studies have been conducted to develop non-biocidal and non-stick FR coatings [43a,35].

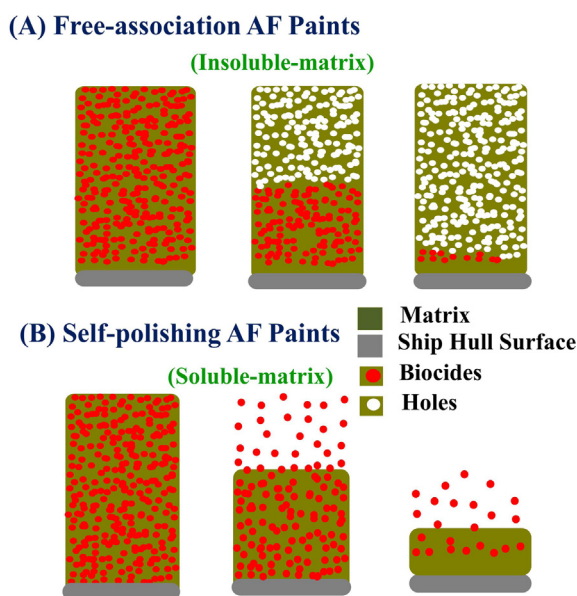


Fig. 3. Mechanism of biocide release from biocidal coatings of ship hull by using (A) insoluble-matrix (free-association) or (B) Soluble matrix (Self-polishing) in AF paints.

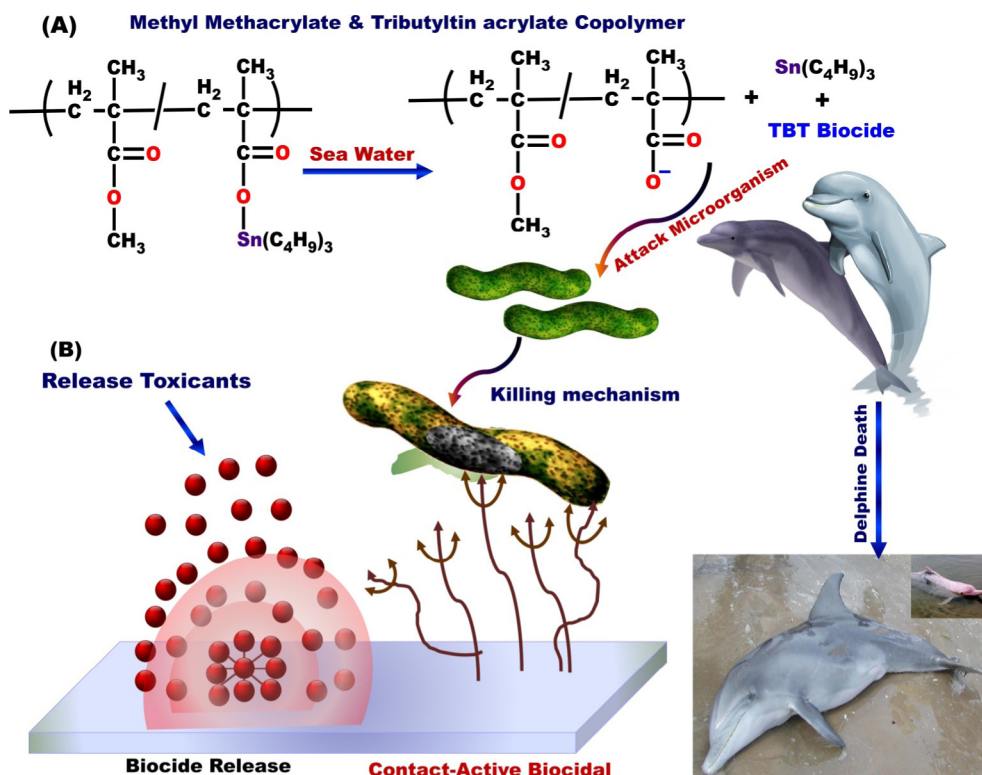


Fig. 4. (A) Leaching of a common copolymer methylmethacrylate and tributyltin acrylate in sea water to release the biocides; (B) Killing mechanism of biocidal AF coatings that prevent fouling adhesion via their toxic effect.

5. Modern approaches to environmental-friendly AF coatings

The functionalities of polymer brushes enabled their potentials in AF paints, especially for the shipping industry. Marine coatings based on polymer brush technology have been developed for FR applications with eco-friendly effects [4]. These polymer brushes inhibit marine biofouling with insertion of specific functional groups, thereby enhancing the anti-adhesion and anti-bacterial non-toxic efficiencies. Three techniques have been utilized in designing AF polymers: (1) degrading or killing of biofoulant; (2) fouling resistance (preventing biofoulant attachments); and (3) fouling release (reducing the strength of biofoulant adhesion) [3]. Prevention of the adsorption protein, cells, or microorganisms is largely affected by controlling polymer structures and coating process techniques. The criteria for non-fouling polymers include the existence of hydrogen bond acceptor and polar functional groups, and the absence of net charge or hydrogen bond donor groups [61]. In this section, we focused on nano-structured functionalities and fidelities, and surface properties of synthetic polymers and their derived nanocomposites that display eco-friendly non-fouling performance. The surface properties are significantly affected by the brushy structure of polymeric coaters and AF operating conditions.

5.1. Fouling-resistant coatings

Amphiphilic polymer nanocomposites are dominant fouling-resistant polymer coatings. These materials possess low polymer–water interfacial energies. The high hydration degree increases the energetic penalty of removing water when bio-foulants attach to a surface; as a result, the surface becomes resistant to protein adsorption and to settlement of fouling organisms [62]. Several hydrophilic polymer nanocomposite coatings, such as polyethylene glycol (PEG), hydrogel, zwitterionic, and hyperbranched polymers, have been developed as marine AF coatings.

5.1.1. PEG-based fouling-resistant nanocomposites

PEGylated materials have been applied because of their strong AF tendency against cell and protein cohesion. PEG is non-toxic, highly hydrophilic, and neutrally charged. PEG presents weak basic ether linkage and reduced interfacial energy with water (5 mJ/m^2), and these characteristics facilitate its good AF performance [63]. Maximizing the surface hydrophilicity and minimizing the attraction forces (caused by formation of hydrogen bonds with water) with fouling community are the mechanistic key issues of PEG [63]. The AF efficiency of PEG with long chains is higher than that of oligo(ethylene-glycol) (OEG)

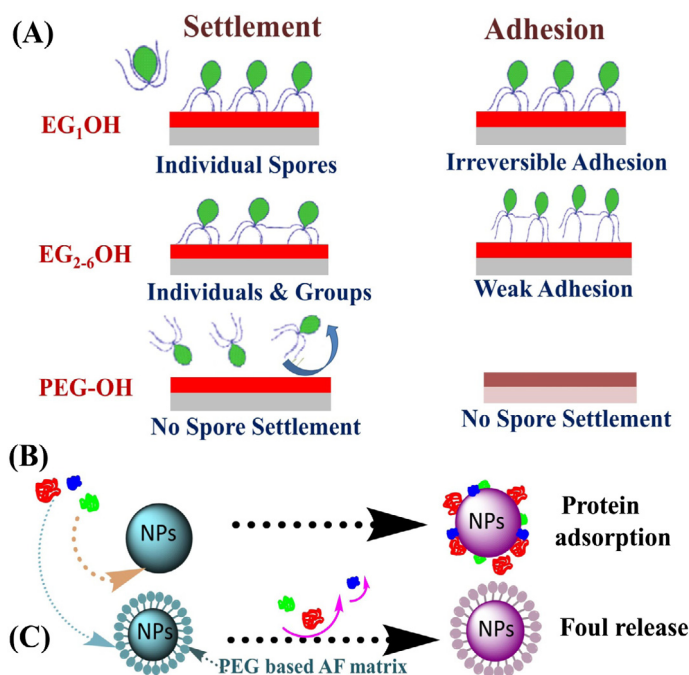


Fig. 5. A Schematic illustrating of the different responses of spores and larvae to OEG and PEG surfaces [64a]. Copy right 2009, reproduced with permission from American chemical society; (b) indicate NPs without AF polymers (PEG) suffer from nonspecific protein adsorption on a macroscopic surface; (C) Attaching AF polymers to NP surfaces allow the particles to resist protein adsorption and prevent fouling adhesion [64b]. Copy right 2015, reproduced with permission from royal society of chemistry.

coatings. PEG prevents spore and larval attachment, and OEG-modified surfaces reduce the adhesion between spores and surfaces; as a result, PEG is easily released through minor hydrodynamic forces (Fig. 5) [64]. PEG coatings suffer from rapid auto-oxidation in the presence of oxygen and transition metal ions, thereby leading to the decomposition of coatings. However, the formation of PEG nanocomposites significantly overcomes this drawback [65]. For example, PEG-ZnO nanocomposites reduce the protein adsorption by 30%. This FR nanocoating is more effective than the silicone hydrogel coating with Ag-polyvinylpyrrolidone that exhibits 28.2% reduction in protein adsorption. Furthermore, most bacteria attached on ZnO-PEG nano-surfaces are eliminated after 4 h of incubation, whereas those on Ag-polyvinylpyrrolidone ones are effectively eliminated after 8 h [65]. PEG-grafted multiwall carbon nanotubes (MWCNTs) nanocomposites have also been synthesized and used as an AF agent enriched with nanofillers to prepare nanohybrid polyethersulfone (PSf) surfaces. Therefore, PEG-g-MWCNTs/PSf nanohybrid systems with effective hydrophilicity and AF performances can be applied to water purification technologies [64]. These nanocomposites, especially with 1.5% PEG-g-MWCNTs, are also promising materials because of their enhanced hydrophilicity, surface area of nanofillers, AF capability, and mechanical strength [64,65].

5.1.2. Hydrogel-based coatings

Hydrogels are composed of hydrophilic polymer networks and are distinguished from solid materials by their high water composition. These hydrogels are porous, three-dimensional network structures that contain 80% water; they are also non-toxic, highly elastic, and inert against bio-macromolecule adhesion that may resist the irreversible protein fouling [65,66]. These key factors inhibit coupling formation between the cement protein and the hydrogel surface, thereby prohibiting fouling attachment. Hydrogels with long PEG chains outperform coatings with short PEG chains because of their capability to prevent the attachment of fouling organisms. Thiol-ene click reaction has been facilitated to prepare PEG hydrogel coatings with different structural compositions, various PEG lengths, vinyl end groups, and thiol cross-linkers [66]. Although hydrogels provide several advantages, such as efficient mass transfer, hydrophilicity, and stimulus- and cell-induced responsiveness, their widespread use is hampered because of their poor mechanical properties and brittleness upon dehydration [61]. Different strategies have also been used to solve these drawbacks; forming hydrogel nanocomposites, such as nanoclay in hydrogel polymers, is the optimum option to create mechanically robust hydrogels (Fig. 6) [66]. Nanoclay, which is composed of inorganic silicate NPs, offers a high interfacial area and improved mechanical and rheological properties because of the reinforced polymer matrix [67]. A hydrogel nanocomposite of carboxybetaine methacrylamide and 2-hydroxyethyl methacrylate and clay NPs exhibits high AF and mechanical characteristics [67]. Furthermore, nanocomposite hydrogels with interpenetrating polymer network structures based on PEG methyl ether methacrylate-modified ZnO NPs and

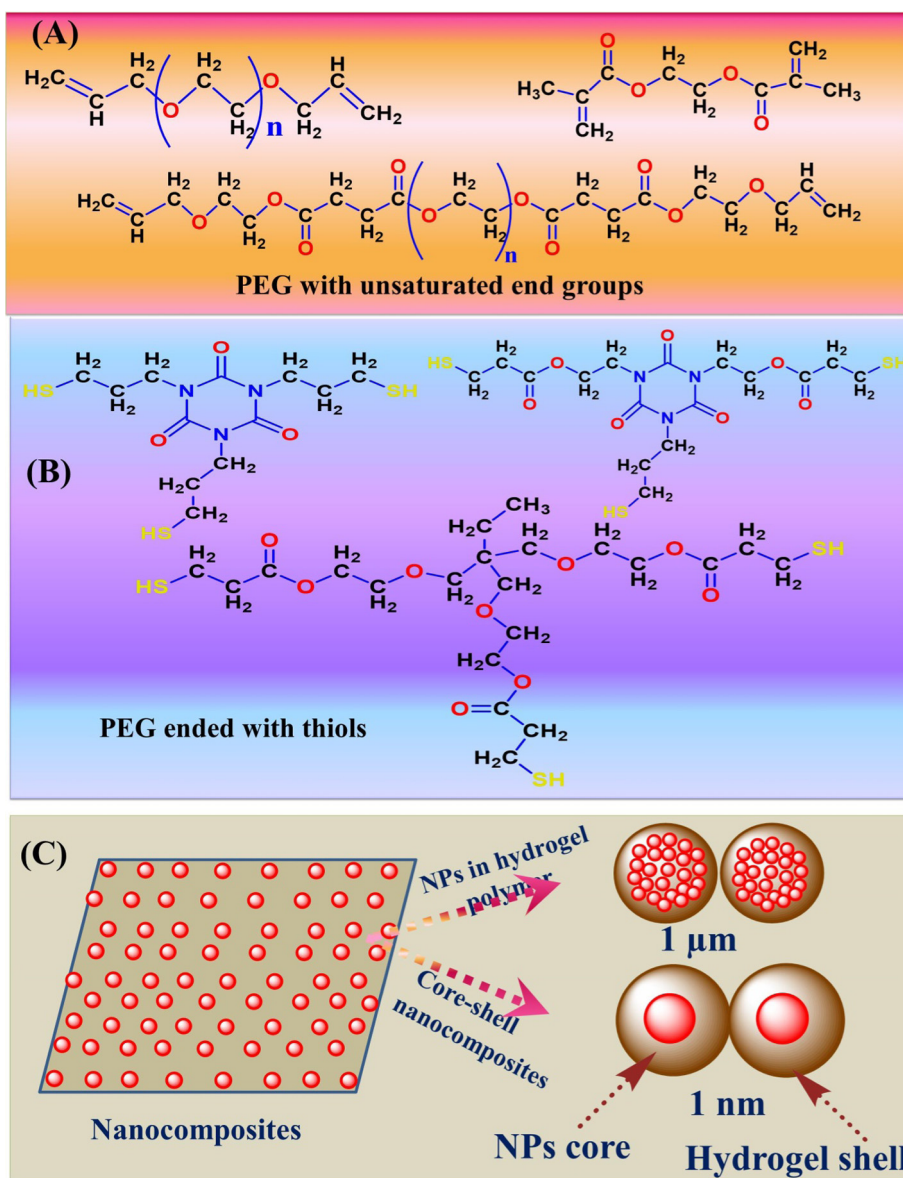


Fig. 6. (a) PEGs with unsaturated end groups; (b) PEG with thiols used for synthesis of hydrogels [4] and (c) Different forms of hydrogel nanocomposites: bulk nanocomposites, particle nanocomposites, and core-shell nanocomposites [67a]. Copy right 2010, reproduced with permission from royal society of chemistry.

4-azidobenzoic agarose exhibit excellent mechanical AF performance with negligible cytotoxicity [67]. These hydrogels manifest high resistance to protein adsorption, cell cohesion, and bacterial settlement; therefore, they present high non-fouling behavior.

5.1.3. Polyzwitterion-based AF nanocomposites

Zwitterionic polymers have been widely explored as a new generation of fouling-resistant materials. These polymers comprise positive and negative charges, which produce more potent and stabilized ionic bonds with water molecules than those created from other hydrophilic materials [62]. These zwitterionic polymers are promising fouling-resistant materials because of their excellent hydration capacity with strong hydrophilicity. Advanced easy-cleaning system has also been designed successfully by tethering a zwitterionic poly(4-(2-sulfoethyl)-1-(4-vinylbenzyl) pyridinium betaine) (PSVBP) onto polyamide (PA) surfaces [68,69]. A PA-grafted-PSVBP exhibits a prominent short-lived fouling prevention and salt-responsive property and becomes unsuitable for long periods. These surfaces can regain self-cleaning abilities by rinsing them with brine [69]. Zwitterionic nanocomposites with controlled NP dispersion and without agglomerations also enhance the fouling

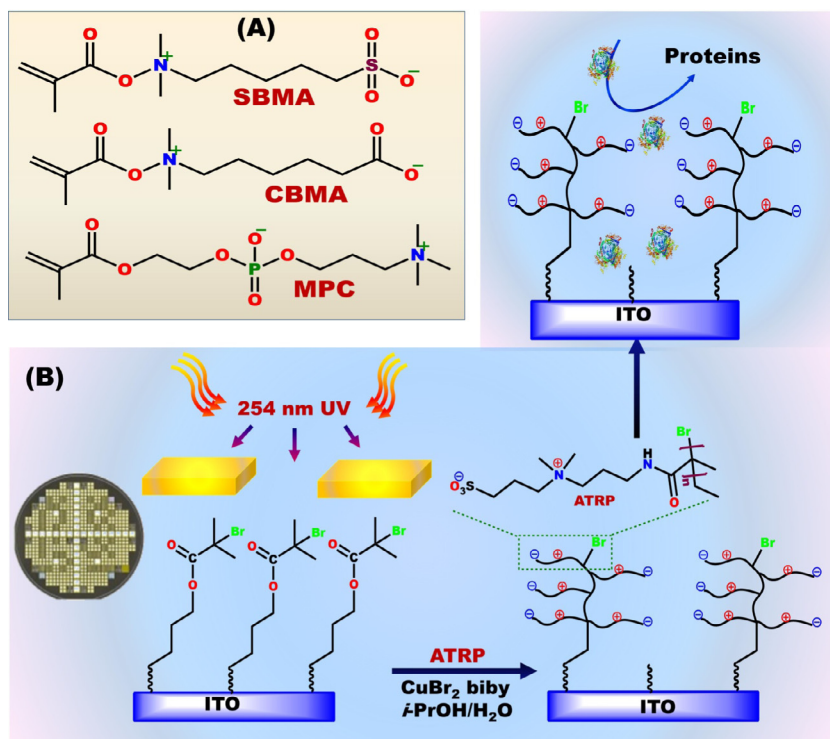
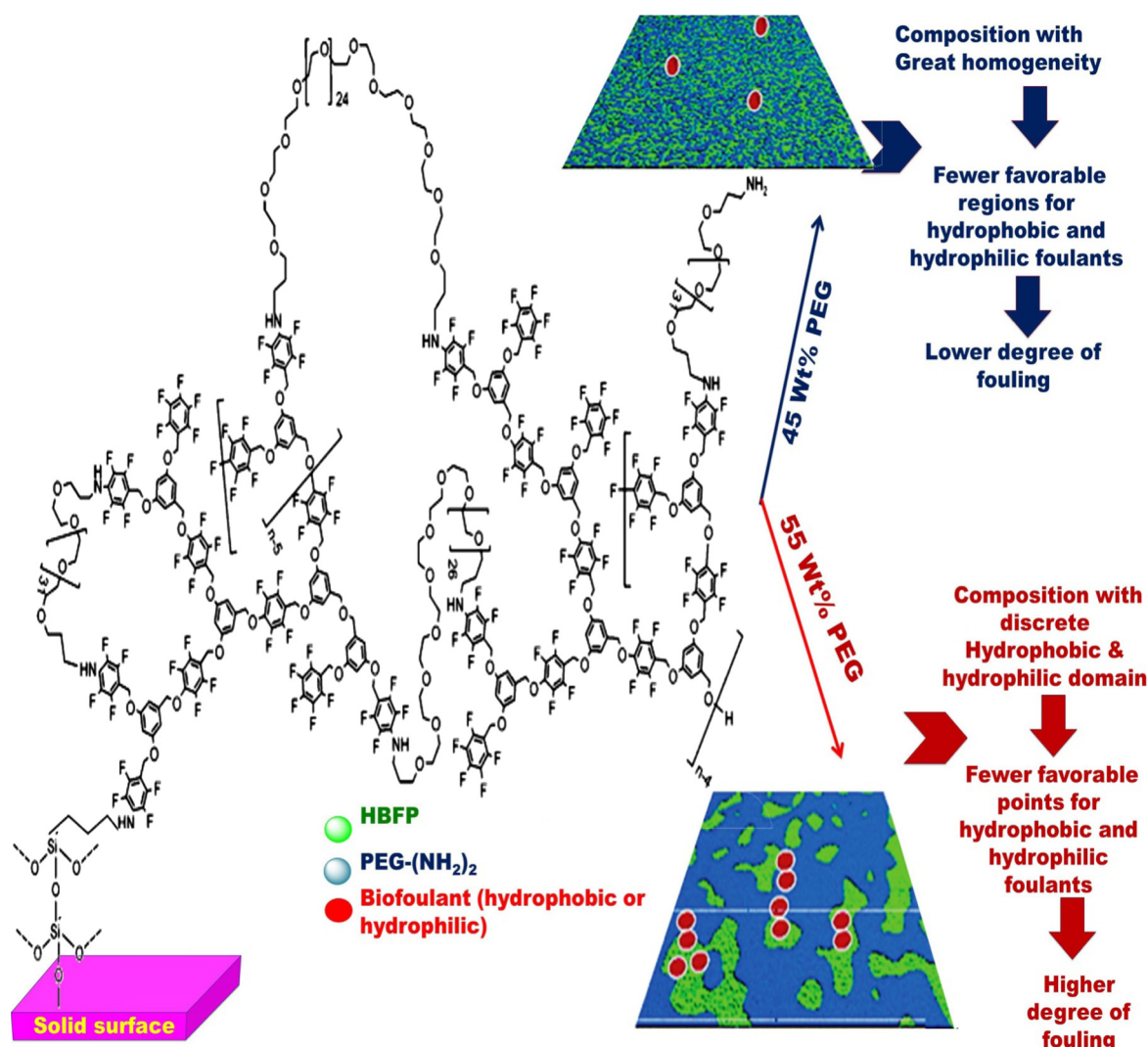


Fig. 7. (a) Chemical structures of some zwitterionic molecules; sulphobetaine methyl acrylate (SBMA), carboxybetaine methyl acrylate (CBMA) and 2-methacryloyloxyethyl phosphorylcholine (MPC) used as AF coatings; (b) Formation and protein adsorption of zwitterionic polymer brushes onto ITO substrates.

resistance of hydrophobic materials (Fig. 7). Zwitterionic polymer brushes are attached to indium tin oxide (ITO) substrates. Photochemically grafted hydroxyl-terminated organic layers serve as an excellent platform to initiate fouling attachment and exhibit excellent AF properties (Fig. 7B) [69c]. Exfoliated montmorillonite nanocomposites with catechol/zwitterionic quaternized polymer were developed, and they present AF properties and resistance against physical damage [69d]. Furthermore, the modification of zwitterionic coating films with SiO₂ NPs can significantly enhance the AF performance [70]. Eco-friendly silver-zwitterion nanocomposites were fabricated, and they exhibit considerable antimicrobial activity and surfaces with anti-adhesion characteristics [70].

5.1.4. Hyperbranched AF nanocomposites

Highly branched coating matrixes with hydrophilic terminals have been extensively investigated to provide biofouling resistance to surfaces. Hyperbranched polymers are likewise advantageous for FR applications because of numerous terminal units, branching density, high solubility, low viscosity, and low volatile organic compound (VOC) [71]. These polymers can form extremely hydrophilic surfaces resembling a hydrogel surface during water contact. Other advantages of these polymers are the simple preparation through single-pot technique and purification; as such, the resulting polymer is a low-cost material [72]. Hyperbranched polyethyleneimine with significant surface topology characteristics shows excellent AF performance and high resistance against non-specific protein adsorption [72c]. Consequently, hyperbranched based nanocomposites provide surfaces with compositional and topographical complexities that hamper any favorable interactions with adhesive biomacromolecular segments secreted by marine organisms. For instance, sericin is a natural hydrophilic protein and contain polar –OH, C=O, and –NH– side chains [65]. This protein is applied as an FR film. AF tests have demonstrated that sericin thin-film composite (TMC) shows a more efficient fouling resistance than commercial composite surfaces do (Scheme 1). This superior performance inevitably results in an increased electrostatic repellency by TMC of sericin toward fouling organisms; as a consequence, foulant molecules are adsorbed on membrane surfaces to a low extent [73]. Deka et al. [73b] demonstrated that hyperbranched polyurethane incorporated with Ag nanocomposites exhibits higher AF properties than linear analogs. The Ag particles cause cellular damage through production of reactive oxygen species. Nanocomposites of catecholic hyperbranched polyglycerol with TiO₂ NPs were developed, and they exhibit significant AF performance [73c]. By applying this design, the protein adsorption decreases with the increase in catechol functionality and with good dispersion of TiO₂ NPs along polymer matrices.



Scheme 1. A Schematic illustrating the structure of polymer networks comprising HBFP and PEG chains. The highest resistance toward protein adsorption is observed for the HBFP-PEG network that contains 45% of PEG [7,161]. Copyright 2005 & 2012, reproduced with permission from American chemical Society.

5.2. Non-stick FR coating approach

Non-stick FR coatings are successful alternatives that inhibit the attachment of marine organisms by providing surfaces with minimum drag resistance and maximum smooth topology [74]. FR coating exhibits promising characteristics as follows (Fig. 8) [74,75]:

- Non-leachant property and no biocide release into the sea.
- Surface with ultra-smooth topology that is conducive to maximum speed, low drag resistance, and minimum fuel consumption.
- Low harmful emissions (such as SO_x, NO_x, and CO₂) to the environment.
- Prolonged fouling prevention and service life (5–10 years) with reduced maintenance and repair costs.
- Extended dry-docking intervals.
- Chemically durable, tin (Sn)-free, copper (Cu)-free, high solid content, and low VOC.

Other advantages include copper-free composition, lower weight than standard AFs, and high solid content [75]. In the early 1970s, FR coatings were manufactured simultaneously with TBT-SPCs. However, the development of FR technology has been limited in few research laboratories [76] because of the efficacy and commercial benefits offered by SPCs. In the

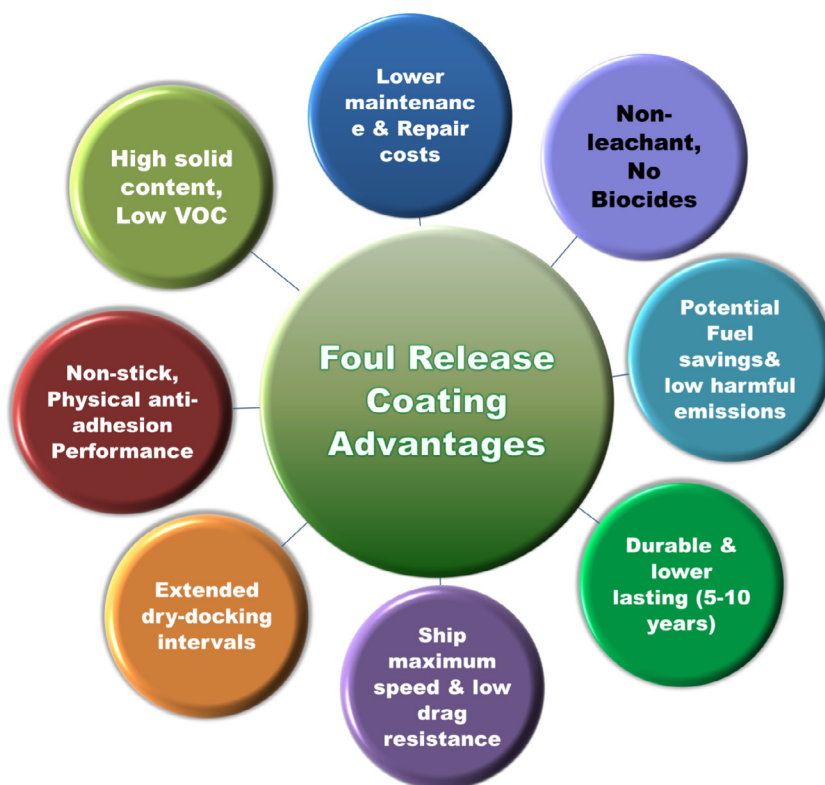


Fig. 8. Economic and ecological advantages of applying FR coatings on ship hulls.

1980s, FR systems were commercially developed because the demands for the rapid delivery of passengers and cargoes have increased and vessels with enhanced design speeds have been developed. This initiative also coincided with the move to ban TBT AF paints. In March 1996, FR coating was first used for a full fast ferry application on a 33-knot aluminum catamaran [77]. With FR coating, the performance of the catamaran improved, with an increase in speed of 2–3 knots under all weather conditions compared with the performance of a newly applied low-copper TBT-free AF system, which was also widely used in those days [74].

5.2.1. Comparative performance of biocidal and FR coatings

The commercialization of micro-topological surfaces for large ships is limited mainly by painting costs and impractical applications [35]. Following the ban of organo-tin compounds by the IMO in 2003, the total sales of FR coatings have increased significantly. A hypothetical complete conversion to FR coatings is estimated to save 70,000,000 tons of copper-based biocides, 6,000,000 tons of booster biocides, and 20,000,000 L every year globally [7,78]. FR coatings possess high volume solid content (70%) compared with SPC (40–50%) and CPD coatings (50–60%); thus, FR coatings exhibit low VOC [7,78]. FR coatings also require one top layer whereas biocidal AF coatings require two or three layers. This requirement of FR coatings lowers the paint consumption, saves cost, and reduces docking time. FR coatings possess long life efficiency (5–10 years) and do not release biocides such that they are unaffected by biocidal legislation. The potential ultra-smooth surface of FR coatings compared with other technologies result in low drag resistance and maximum fuel savings [79]. The fuel consumption of tin-free SPC coatings and FR coatings were compared through application on the hulls of two vessels by Corbett et al. [80]. The results illustrated that FR coatings decrease speed-adjusted fuel oil consumption by 22%. If similar fuel efficacies are achieved in the international fleet, then the annual fuel consumption, fuel expenditure, and carbon dioxide emission are expected to decrease by 16,000,000 metric tons, \$8,800,000,000, and nearly 49,000,000 metric tons, respectively [7,80]. Data on AF system prices are not easily accessible because coating manufacturers are unwilling to openly share such information. The total cost accompanied with biocidal or FR coatings depends on the liquid paint, off-hire for the vessel, dry-dock hire, hull cleaning, surface preparation, and roughness costs [80]. Roughness increases by 10%, 4.8%, and 1.4% for hybrid SPC, SPC, and FRC, respectively, after a 5-year service life because of fouling. Eliasson [81] indicated that FR coatings are less costly than biocidal AF paints even after 5 or 10 years. FR terminology, which is awarded by fluoro and silicon polymers, is also applied by minimizing the adhesion strength between material surfaces and fouling organisms. Thus, organisms can be easily removed through mechanical cleaning or hydrodynamic stress during shipping.

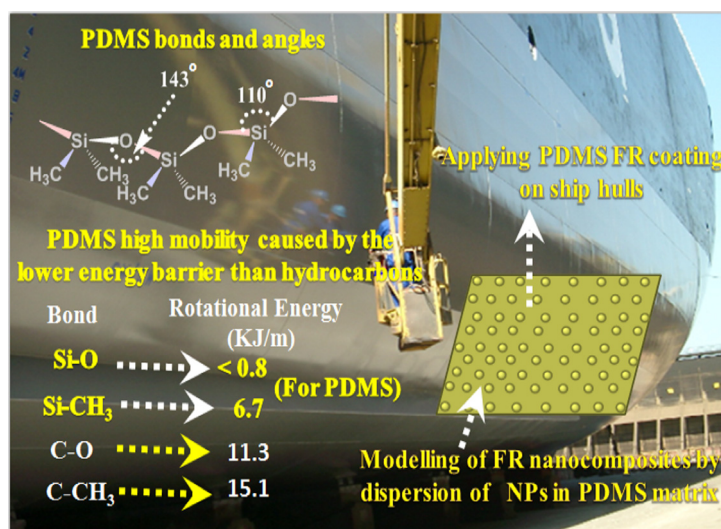


Fig. 9. Application of PDMS FR coatings and inside (1) Bond angles and bonds of siloxane structure; (2) low rotational energy barrier of Si-O and Si-CH₃ bonds in PDMS as compared to C-O and C-CH₃ bonds in hydrocarbons (which explain the high mobility and lower T_g and thus the high FR performance of PDMS) and (3) on the right hand nanocomposite formation through dispersing NPs in PDMS matrix and then applying on ship hulls.

5.2.2. Fluoropolymeric FR coatings

Fluoropolymers form non-porous and smooth surfaces with minimized surface tension and good anti-adhesion performance toward fouling organisms [82,83]. These substances also create a weak interface with marine adhesives, and fouling attachment can be simply disrupted by shear stress. Fluoropolymers are hard, glassy materials with glass transition temperature (T_g) that exceeds room temperature (RT). These polymers are applied to thin layers (75 μm) to produce non-stick FR coatings (Fig. 7). However, these materials exhibit limited mobility because fluorine atoms form stiff molecular structure and thus inhibit bending at the back bone chain [83]. In addition, high critical stress is necessary to induce failure on the adhesive–substrate joint because fluoropolymers yield a higher bulk modulus than elastomers do [84]. Therefore, fouling organisms that accumulate on surfaces are not easily released. Thin coatings require very high removal forces because stress and force are inversely related to thickness. As an attempt for modification, fluorine compounds can be mixed as a polymer moiety or as curing agent to the silicone coatings. These compounds possess lower surface tension than silicone; this condition allows an interfacial immigration of fluorinated moiety to the surface and reduction of the coating surface tension, thereby providing high chemical stability in water [85,86]. However, fluorinated silicone coating induces fragile layers, which are prone to splits and cracks. The drawbacks in applying fluoropolymers as FR coatings have prompted modern research toward elastomeric silicone FR coatings.

5.2.3. Silicone compounds in FR coatings

Silicone derived its name because of the structural resemblance between R_2SiO and R_2CO of ketones. Structural studies on bond lengths have indicated that the lengths of the siloxane backbone bond and carbon–carbon bond are 1.64 and 1.53 Å, respectively; thus, the longer siloxane bond rotates simply. The siloxane angles are 143° and 110° that correspond to Si–O–Si and O–Si–O, respectively; these angles are higher than the tetrahedral bond angle of 109.28° in hydrocarbon chains (Fig. 9) [87]. The rotational energy for free rotation of Si–O and Si–CH₃ in PDMS is lower than that for C–O and C–CH₃ in hydrocarbon polymers. This condition allows high structural mobility and low T_g for the silicone backbone (Fig. 9). The PDMS bond angles and the existence of an alpha-helix configuration form an open structure and flexible bonds that rotate through a 180° state and a low rotational barrier. As a result, isotropic protection that accounts for the lowest T_g (around –120 °C) of any other polymer and low interfacial free energy (around 20–24 mJ/m^2) is provided [88]. Another major factor that contributes to the increased flexibility of PDMS is that O atoms in the skeletal siloxane bonds are not encumbered by any side chains. Moreover, all silicone backbone bonds yield a very low torsional barrier for free rotation, which accounts for the mobile configuration. The linear PDMS is liquid at RT because its T_g is less than its RT. Methyl groups (–CH₃) are characterized by very intense σ bonds; as such, these groups protect the –Si–O– groups, which are reactive because of their polar nature [89]. Organo-silicon polymers, particularly PDMS, represent a marked niche among specialty (co)polymers. These advantages provide PDMS with superior fouling anti-adhesion characteristics, surface inertness, hydrophobicity, high heat resistance, and excellent insulation property (Fig. 10) [90]. PDMS also presents viable options in several hazardous and polluting industries because of their environmental-friendly nature. With regard to market value, silicone products reached US\$ 13.5 billion in 2010 and are expected to reach US\$ 17.2 billion in 2017 because of their extraordinary properties [90–92]. PDMS can inhibit the attachment of fouling functional units and release their attachments even at low sailing speed [93].

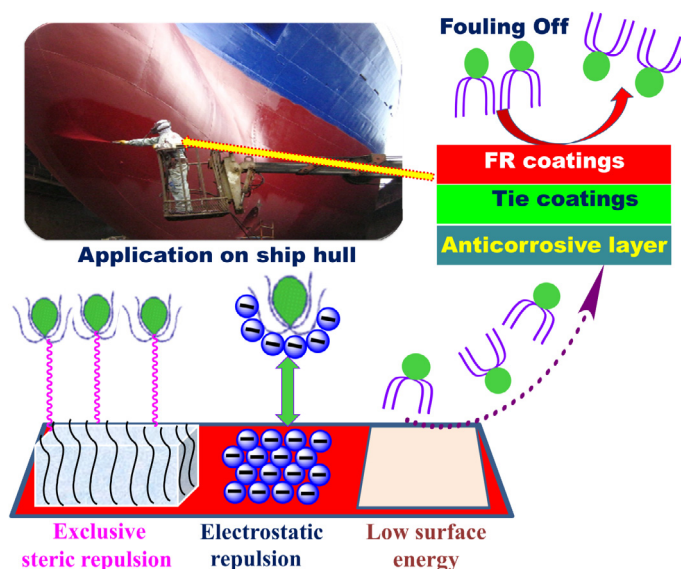


Fig. 10. FR coating mechanism which prevents the attachment of fouling organisms via three parameters exclusive steric repulsion, electrostatic repulsion and low surface energy.

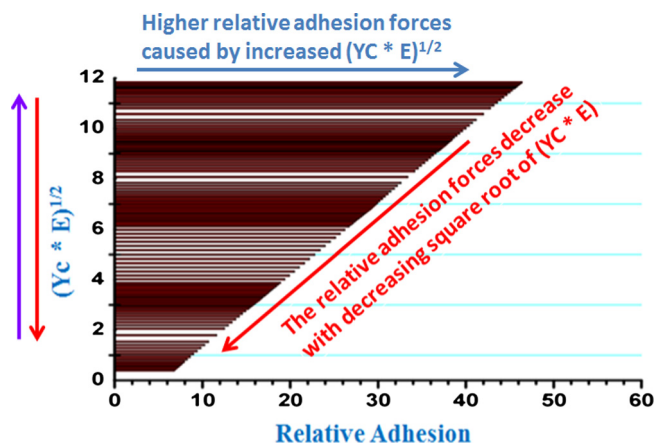


Fig. 11. Relative fouling adhesion decreased (enhanced FR performance) with decreasing the square root of surface tension and elastic modulus $((Yc * E)^{1/2})$ [11]. Copyright 2000, reproduced with permission from Taylor & Francis.

Kohl and Singer [89b] tested the resistance of silicone coatings to marine fouling and found that fouling capability is correlated with the square root of the result of multiplying surface tension (Yc) in the modulus of elasticity (E) (Fig. 11). The relative adhesion forces decrease with the decrease in either surface tension or elastic modulus. Minimum adhesion properties coinciding with the lowest E have also been investigated; results indicate that the E is also a major factor in silicone FR surfaces [89b].

Although PDMS has been considered non-biodegradable by fouling organisms, its biodegradability data are insufficient and the biodegradation by living organisms have been poorly studied [89,94]. PDMS biodegradability mechanism gives rise to the formation of dimethylsilanediol, carbon dioxide, and inorganic silicate as illustrated in Fig. 12. Another major factor is that increasing the PDMS molecular weight reduces the interfacial tension of the surface and thus enhances hydrophobicity and self-cleaning performance of the FR paints [95].

6. Recent FR silicone nanocomposite coatings

Innovation of advanced materials, especially the functional hybrid nanocomposites, has become an urgent need to meet the promise of improved and often unparalleled characteristics at a reduced cost. Hybrid organic–inorganic nanocomposites provide the opportunity to combine the properties of both materials and generate advanced chemical, physical, and biolog-

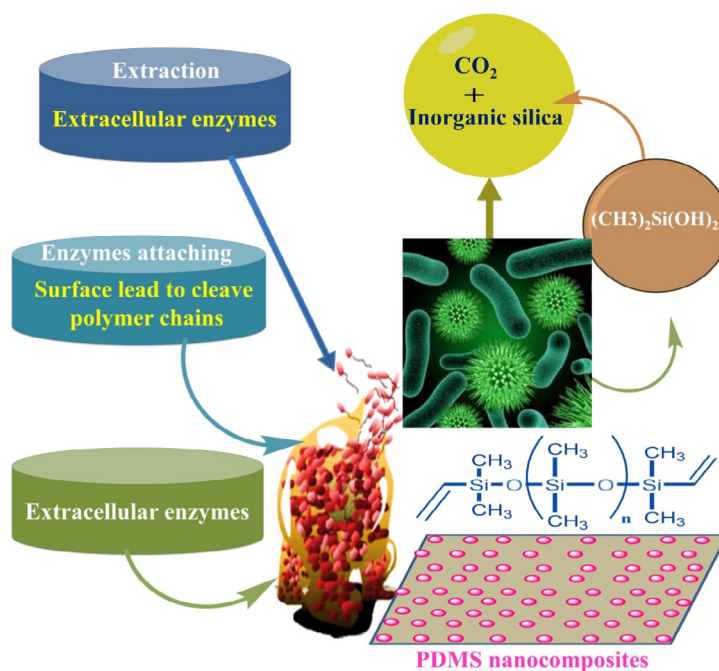


Fig. 12. The biodegradation mechanism of PDMS nanocomposites by the action of microorganisms.

ical properties. These engineered nanomaterials have resulted in a technological leap in various fields, such as electronics, biological applications, pharmaceuticals, renewable energies, and coatings [96]. The availability of structure–property relationships in AF coatings through good dispersion of controlled NPs in polymer matrices can ameliorate bio-static, thermal, anticorrosive, self-cleaning, and AF properties [97]. After mixing NPs with large polymers, necklace-like structures of trapped NPs into polymer coils are obtained with altered rheological behavior [98]. At high NP concentrations, a few NPs settle in the coil periphery and form bridge aggregates through sharing; these aggregates can weaken the coil junctions [96]. This problem is due to the electrostatic repulsion between the adsorbed and free particles. The opposite situation, in case of well dispersion, is reported [98c].

PDMS-based nanocomposites have been extensively investigated because of their facile preparation and environmental stability. Inorganic nanofillers are one of the promising solutions for superior PDMS-based nanocomposite coatings. Various NPs, such as SiO_2 , Al_2O_3 , Fe_3O_4 , ZrO_2 , and TiO_2 , have been introduced to modify FR membranes based on reinforcing the physical, chemical, and mechanical performance [99,100]. However, an occurrence of a swap between self-cleaning of PDMS paints of vessel bottom surfaces and durability controls the morphology, size, and percentage of nanofiller insertion. Biocidal AF paints function by possessing an acid binder and dissolved biocides, which undergo leaching and releasing into sea water to kill the fouling organisms (Fig. 3). In turn, silicone-based FR paints exhibit non-leachant property and are thus environmentally controversial. These silicone-based FR paints reinforce the bonds of fouling attachments, thereby releasing the attached organisms back to the sea without killing them (Fig. 10). Bactericide-hydrophobic nanofillers, such as MWCNT, natural sepiolite, and organo-modified montmorillonite, have also been incorporated in PDMS FR coatings [101]. Insertion of hydrophobic nanofillers in PDMS increases CA, decreases roughness, reduces the surface free energy, minimizes elastic modulus, and improves the self-cleaning characteristics. Incorporating a small amount of MWCNT (less than 0.1 wt.%) within the silicone coatings enhances the fouling release performance against fouling organisms without considerably changing the bulk properties [102]. Insertion of natural sepiolite $\text{Si}_{12}\text{O}_{30}\text{Mg}_8(\text{OH})_4(\text{H}_2\text{O})_4 \cdot 8\text{H}_2\text{O}$ NPs in commercial vinyl terminated PDMS (Dow Corning Sylgard® 184) reduces the modulus of elasticity and the relative adhesion of fouling communities. The high surface area of sepiolite is attributed to its micropores, channels, fine particle size (5–10 nm in diameter), and fibrous nature [101,102]. This clay structurally forms blocks and channels extending in the fiber direction. Some Si–O–Si groups in the silica tetrahedral sheets are present as silanol groups (Si–OH) as a result of imperfections. These groups also facilitate the covalent modification of these particles. However, the FR performance of the filled nanocomposites only affected anti-adhesion of definite biofouling organisms such as zoospores and *Ulva* [103]. Although, the initial hydrophobicity was changed and the adhesive tendency of adult barnacles was reduced by 50% after addition of 0.05% MWCNTs, the tailored PDMS/MWCNT nanocomposites did not affect the durability characteristics [104]. The strong CH– π interactions between silicone methyl side chains, which render mobile molecular configuration and the π -electron-rich surface of MWCNT, are responsible for the unexpected behavior. Modeling of fluorinated carbon nanotubes (CNTs) for silicone-based FR coatings is a modern

Table 2

Contact angle measurements and root mean square roughness (RMS) determinations of water repellent surfaces with dual (nano-scale and micro-scale, coatings 1 and 2) and single (nanoscale, coating 3) roughness. Adopted from Ref. [110].

	Coating 1 ^a	Coating 2 ^a	Coating 3 ^a
Contact angle (Advancing/receding)	169°/160°	155°/145°	169°/160°
RMS (Scan rate 5 μm)	309 nm	482 nm	374 nm
RMS (Scan rate 20 μm)	1117 nm	1463 nm	607 nm

strategy for high FR performance and improved self-cleaning capability [105,106]. Irani et al. [107] investigated PDMSs filled with MWCNTs and fluorinated MWCNTs at different concentrations of 0.05, 0.1, and 0.2 wt.%, and they found that these materials can be successfully fabricated and used for FR treatment. Both structural types of nanotubes alter the nanocomposite surface characteristics; however, the fluorinated MWCNTs/silicone composite model reveals superior FR performance in ship hull. These studies indicated that the fluorinated MWCNTs/silicone composite may decrease the strengths of pseudo-barnacle attachments by 67% compared with pure MWCNTs/silicone nanocomposite models, which exhibit reduction of only 47% [107,108].

However, single-wall CNT, graphene, and their modified models are promising in the reinforcement of self-cleaning, fouling anti-adhesion and mechanical characteristics of the designed FR-based PDMS nanocomposites [104]. Other terminology involves insertion of low surface tension fluid such as oils which undergo air interference migration and create weak surface bonds resulting in anti-adhesion capability [109]. The additive concentrations should be studied carefully because brittleness, cracks, and low durability result when some of these materials are added at high concentrations [110]. Lowered hydrophobicity is regained after immersion in water, indicating that the silicon-based compounds have renewable properties. Webster et al. [111] extensively investigated self-stratified cross-linked polysiloxane–polyurethane coatings and reported a considerable increase in the critical stress removal of some species of acorn barnacles [112]. PDMS–poly (caprolactone) surfaces were reported to decrease the free energy of the modified coatings from 21 mJ/m^2 to 16 mJ/m^2 which improve biofouling detachments.

Several nanocomposites possess superhydrophobic characteristics i.e. a CA of more than 150° is applicable for FR coatings of ship hulls [113]. The superhydrophobic characteristics of three polysiloxane-reinforced silica fumed NPs composite paints with different structural compositions were tested by Scardino et al. [114]. The three nanocomposite coatings had different roughness measurements where an overlapped micro-nano roughness was recorded for the first and second coating while the third coating records only with nano-roughness (Table 2). Three attachments of different biofouling organisms; motile (*Ulva spores*), subsequently motile (*Diatoms*) and non-motile (*polysiphonia spores*) were assessed after considering the same CA for the three coatings. Results clarified the superior FR performance of the third coating. The study reflected that not only the surface hydrophobicity is the dominating factor for fouling prevention but also the topology of the surface, roughness length scale, and the percentage of entrapped air at the coatings.

Fabricating superhydrophilic/superhydrophobic patterns without toxicants is very challenging. TiO_2 –PDMS composite films have been fabricated by sol-gel method followed by curing with UV light. These composite films could recover their initial wettability. Superhydrophobic TiO_2 nanowire coatings have been reported using the low-surface energy PDMS modification which can resist photocatalytic breakdown caused by the embedded TiO_2 NPs [115,116]. Sankar et al. [13,117] reported the FR performance of the CuO, cetyl trimethyl ammonium bromide (CTAB)–CuO and ZnO nanofillers when incorporated in PDMS coatings [118]. The superior AF properties were approved for CTAB–CuO with low concentration (0.1 wt.%) NPs because of the improved hydrophobicity, nanofiller dispersion, surface smoothness, and antibacterial properties [13,118,119]. Several parameters can influence the AF performance of the utilized NPs such as utilization of capping agents, ultrasonication, nanofiller dispersion, particle size, and morphology [119]. Two-dimensional ZnO-containing porous boron-carbon-nitrogen sheets dispersed in PDMS form a stable coating over aluminum alloys with water CA of 157.6° , which demonstrates oil fouling-resistant superhydrophobicity and self-cleaning. Previous studies indicated that the coating remains stable under high humidity at a thickness of 30–40 μm without cracking when PDMS matrix is incorporated with SiO_2 NPs [120]. A series of non-toxic nanostructured coatings based on PDMS–polyurea/nanoclay segmented copolymers has been investigated, and these coatings exhibit high mechanical strength coupled with enhanced hydrophobicity, self-cleaning, and FR characteristics [121].

Several metal oxide NPs of TiO_2 , Cu_2O , and Ag exhibit unique bactericidal activities caused by their chemical properties, quantum confinement effects, and large surface area-to-volume ratios. Selim et al. investigated the real AF applicability of the ecofriendly silicone filled Ag, PDMS/ Cu_2O , and PDMS/ TiO_2 nanocomposites with different concentrations ranging from 0.01% to 5% nanofillers. PDMS/Ag (0.1%) and PDMS/ Cu_2O (0.1%) nanocomposites showed enhanced properties caused by the good dispersion and complete disagglomeration, which improved surface roughness, and hydrophobic wettability [122,123]. The PDMS/Ag nanocomposites (0.1% nanofillers) exerted the most profound self-cleaning and FR effects after comparison with many approved systems; (1) the hydrosilated cured Sylgard[®] 184 (Dow Corning Co.) and the condensed cured RTV11 (Dow Corning Co.) [124–126]; (2) Sylgard 184/MWCNT [125]; and (3) PDMS nano-filled with 0.1% cubic

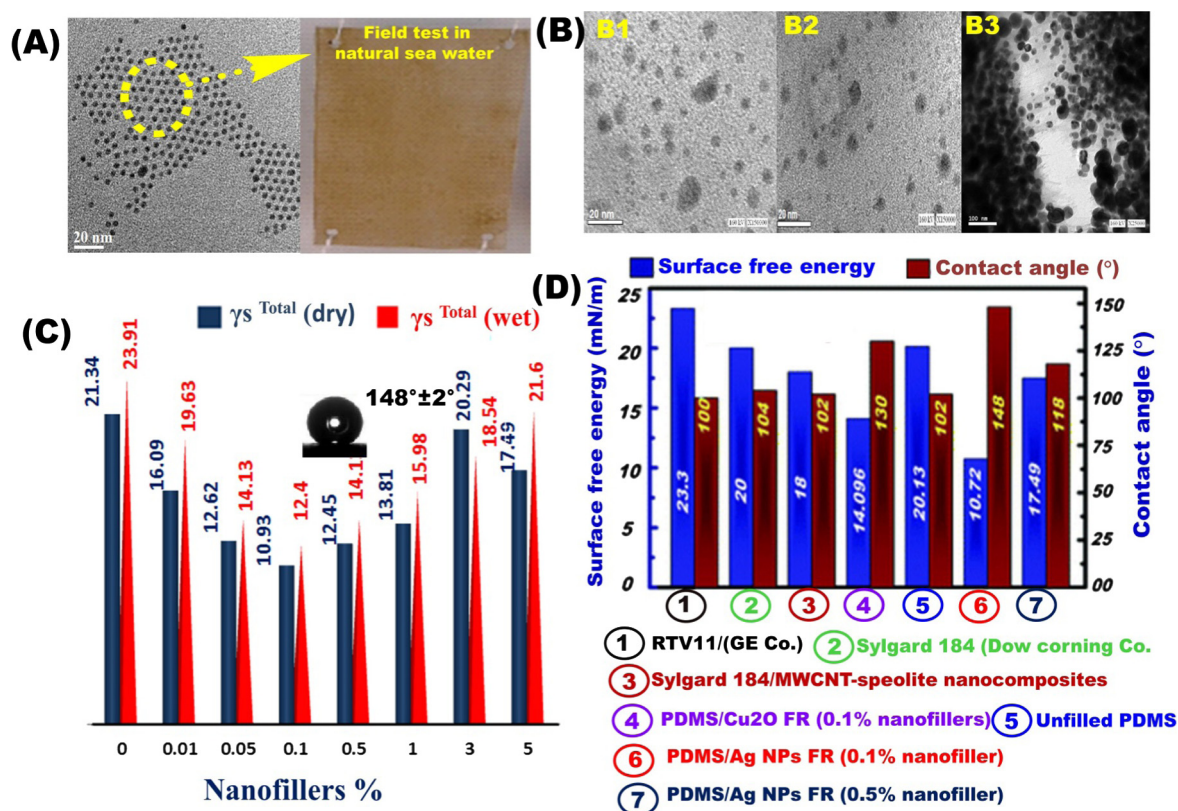


Fig. 13. (A) Studying the effect of different nanofiller loadings of Ag in the silicone nanocomposites; the TEM images of Ag silicone nanocomposite explain the value of inserting well dispersed NPs (B1&B2) and the problems of agglomeration NPs (B3) on the wetting characteristics and FR performance; (C) the contact angles and surface free energy measurements of different Ag nanofiller concentrations in the modeled FR nanocomposites for comparable purposes; (d) a comparable study of the tailored PDMS/Ag nanocomposites and other commercial FR paints [119]. Copyright 2015, reproduced with permission from royal society of chemistry.

Cu₂O coating [118,119]. Among the four FR models, PDMS filled with 0.1% Ag achieved the maximum hydrophobic characteristics and the minimum roughness and interfacial tension [125,126]. These properties were confirmed by applying the nanocomposites in natural waters of the Red Sea to be a promising environment-friendly material (Fig. 13). The exposure of Ag NPs to the aqueous environment promotes the significant release of Ag⁺ ions [127,128].

Selim et al. [129] developed smart photo-induced silicone/spherical single crystal TiO₂ nanocomposites as a novel terminology for solar light boosted FR paints of vessel bottoms. They developed photo-smart silicone nanocomposites aimed to enhance the durability and self-cleaning AF performance through green pathway based on the perfect usage of solar light irradiation. The results indicated the considerable distribution and complete disagglomeration of TiO₂ NPs in the silicone matrix at controlled low concentrations of up to 0.5% loadings [125]. Incorporating spherical single-crystal TiO₂ NPs remarkably influences the coating roughness, wettability, and FR performance when these materials are pulsed by UV–vis irradiation. Self-cleaning through increased superhydrophilicity is one pathway for FR coating (Fig. 14) [129,130]. We compared the static CA measurements for Sylgard®184 silicone film as a control sample [124], PDMS/nanorod TiO₂ composites [130], PDMS/anatase spherical TiO₂ nanocomposite [131], and our newly designed PDMS/spherical single crystal rutile TiO₂ nanocomposite with its superior concentration (0.5% TiO₂ NPs). We found that the performance of FR is superior to that of the tailored nanocomposites. Comparative results based on CA varied after pulsation by UV irradiation; CA remains unchanged after UV irradiation for unfilled Sylgard®184 but decreases from 104° for Sylgard®184 to 77° for PDMS/nanorods TiO₂ composites and to 80° for PDMS/anatase spherical TiO₂ nanocomposite [131]. However, the CA in our developed model (0.5%) decreased to 10° after UV irradiation, indicating superhydrophilicity and increased self-cleaning capability and thus higher AF performance (Fig. 14c). These results were confirmed by the free energy of hydration (ΔG_{st}), which exhibits superhydrophilicity (-144.49 mJ/m^2) and inhibits fouling cohesion on the ship hull surface. These results were confirmed by biological assays and field exposure tests, which revealed the improved durability and self-fouling-releasing performance. The tailored nanocoatings are ecofriendly and are possible economic long term solutions which could replace the harmful bio-cidal AF materials. Future efforts should be spent to design novel FR nanocomposite coatings for economic savings and prolonging longevity.

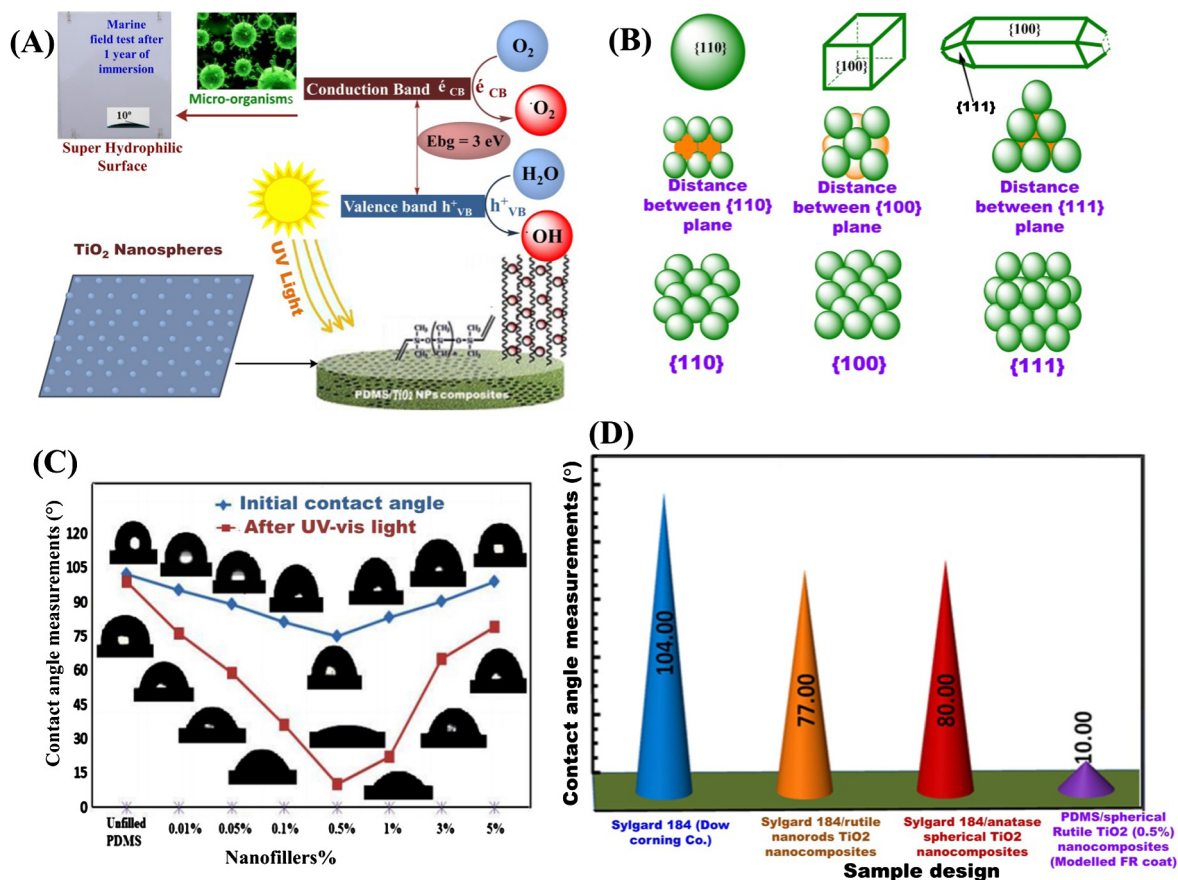


Fig. 14. (a) utilization of solar light irradiation for boosting the self-cleaning FR performance of silicone/TiO₂ nanocomposites; (b) the effect of controlled design of the particle size morphology is carried through a comparison of the different crystal structures of the {110}, {111}, and {100} faces of TiO₂ NPs which directly affect the photocatalytic performance and is central in designing solar-spectrum-boosted self-cleaning; (c) a comparison between contact angle measurements of a define nanocomposite at different loading concentrations of the nanofillers indicate the effectiveness of well dispersion of the NPs; (d) Comparable survey of the most recently developed superhydrophilic self-cleaning FR nanocomposites and other sounded FR coatings [125]. Copyright 2016, reproduced with permission from Elsevier.

7. FR nanocomposites based on the surface wettability

CA (θ) is commonly used to express the wetting behavior of a solid surface and is highly significant in all solid-liquid-fluid interfacial phenomena. Wettability is also expressed by the surface free energy of solid surfaces. The FR technology is based on binary eco-friendly modes of actions as follows:

- “Non-stick” characteristics through ultra-smooth surface and reduced surface free energy, and
- “Release of fouling” action through decreased Young’s modulus.

In 1805, Young [131b] first described the intrinsic CA of a liquid droplet on an ideally flat and homogenous solid surface (Fig. 16a). Young’s equation for computing interfacial surface tension of a solid is as follows:

$$\gamma_{SV} = \gamma_{SL} + \gamma_{LV} \cos \theta_Y, \quad (1)$$

where θ_Y is Young’s CA. γ_{SV} , γ_{SL} , and γ_{LV} refer to the interfacial surface tensions with V, L, and S as vapor, liquid, and solid, respectively. The attachment of marine community has been historically related to critical surface free energy (γ_c), as introduced by Fox-Zisman’s theory [132]. This theory is used to determine γ_c , for any substance with surface energy of less than 72 mJ/m², by using CA data. This approach is suitable for polymer-liquid systems. This approach is related to the surface tension of the used liquid γ_L , where γ_c is obtained through determining the CA of different liquids on a solid substrate. A plot between the angle cosine and γ_L of the liquids assuming $\theta = 0^\circ$ (the best linear fit) [133].

$$\gamma_c = \gamma_L (\cos \theta) = \gamma_s \quad \text{where } \theta = 0^\circ \text{ and } \gamma_s \text{ is the solid surface tension}$$

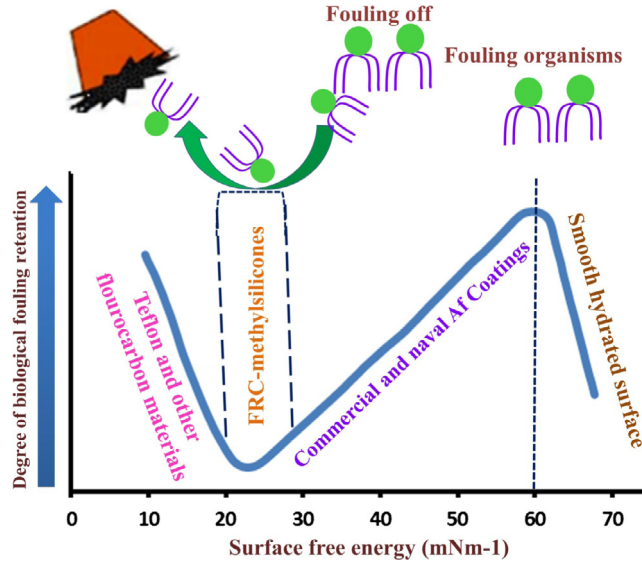


Fig. 15. The Baier curve of adhesion strength vs. surface energy applied in AF paints where Baier curve minimum values improve biofouling detachments [128]. Copyright (2006) with permission from Springer.

$$\gamma_s \geq \gamma_c \quad \text{for apolar liquid } (\theta = 1^\circ)$$

This approach is still debated because the surface energy depends on the nature of the utilized solvents and γ_c only is equal to γ_s for total polar surfaces and liquids; these issues limit its application [133,134]. According to Fox–Zisman's theory and using Owens–Wendt–Rabel–Kaelble's method (also called the geometric mean method), the total surface free energy (γ_s^{total}) of the prepared films and nanocomposites can be determined. This approach suggests that the γ_s^{total} is equal to the sum of a dispersive and a polar components. γ_s^{total} can be calculated from the CA (θ) measurements of different solvents, such as H₂O and diiodomethane [135]. The following equations were used:

$$(1 + \cos \theta_L) \gamma_L = 2[\sqrt{\gamma_L^D \gamma_s^D} + \sqrt{\gamma_L^P \gamma_s^P}] \quad (2)$$

$$\gamma_s^{\text{total}} = \gamma_s^D + \gamma_s^P \quad (3)$$

where θ_L and γ_L represent the CAs and free energy determinations of the utilized solvents, respectively. γ_L^D and γ_s^D represent the dispersion surface tension of the testing liquids and surfaces, and γ_L^P and γ_s^P correspond to the polar surface tension of the testing liquids and surfaces, respectively.

Baier [136,137] investigated the mechanism of microfouling attachment and was first reported in 1968 based on the relationship between surface tension and the strength of attachments. Baier's curve exhibits the minimum attachment strength from 20 mJ/m² to 25 mJ/m², which is caused by forming weak boundary layers between the coating surface and the adhesives of fouling strains (Fig. 15). A lower tendency of fouling strains to attach to ship hulls was reported for hydrophobic coatings with minimum surface tension. The weakly attached fouling organisms can be easily eliminated through the movement of vessels [137]. Three non-stick FR nanocoating techniques toward fouling organisms and their environmental relevance were extensively addressed in this review; these approaches are (i) superhydrophobic, (ii) UV-induced superhydrophilic, and (iii) amphiphilic modes. Owing to the unique properties of graphene-based materials, we introduced a new prospective avenue toward their application in self-cleaning FR nanocomposites for maritime navigation below.

7.1. Hydrophobic/superhydrophobic FR nanocomposite paints

With their wide applications and unparalleled easy-cleaning, non-stick, and ecofriendly characteristics, water repellent (superhydrophobic or lotus effect) surfaces gained worldwide attention [138]. The lotus effect is common in natural environments, such as lotus plants, rice plants, and butterfly wings, which are characterized by their high hydrophobicity and self-cleaning surfaces [139]. Ultra-water-repellent surfaces with water-CA of >150° and where water droplets continuously roll at low-contact-angle hysteresis of <5°, have been extensively investigated. These surfaces are related to wettability mechanism and correlated with surface roughness and surface free energy (Fig. 16). Methyl side chains and the fluorine atoms are not the requirements for lotus effect of natural surfaces and subsequently the very low free energy is unnecessary for water repellency in nature. However, controlling the surface morphology and topology from micro- to nanoscale is the

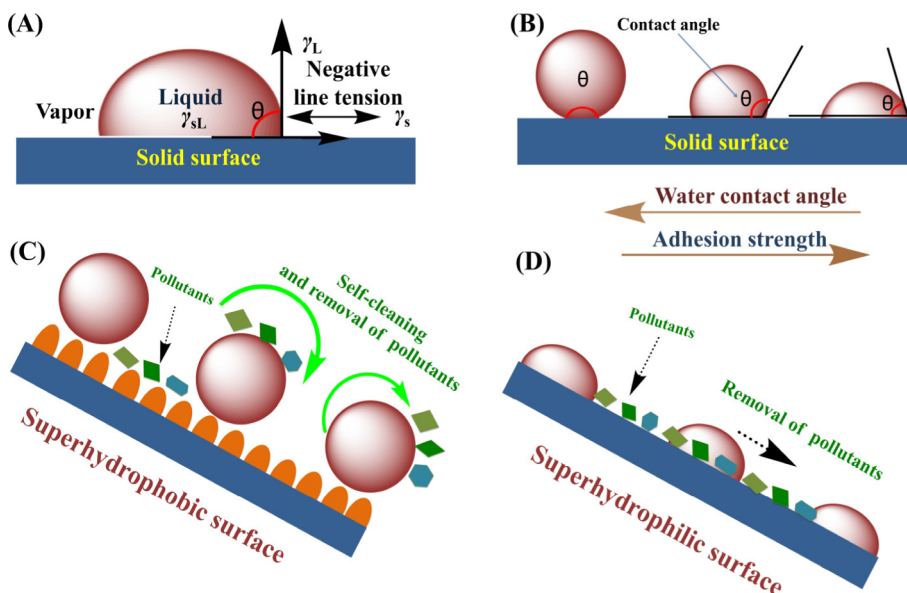


Fig. 16. (a) Young's contact regime for water droplets with a smooth surface; (b) Schematic illustration of effect of the water contact angle on the strength of adhesion; (c) indicating self-cleaning of the wetted surfaces via super-hydrophobic and (d) via superhydrophilic phenomena.

key factor for these natural superhydrophobic characteristics. Lotus effect surfaces can be produced via two facile strategies: (1) designing ultra-smooth surfaces by using low surface tension materials; and (2) designing rough surfaces modified with low-surface-energy materials. Superhydrophobic surfaces have been investigated in terms of biofouling prevention, decreasing friction resistance, anti-corrosion performance, coating easy-cleaning, and snow anti-sticking [140]. Several synthetic techniques such as sol-gel technique, self-assembly, electrochemical deposition, and plasma treatment were used to design self-cleaning materials with lotus effect performance [141]. These surfaces possess surface free energies near the Baier's minimum. These energies decrease bacterial attachment on ship hulls. Furthermore, ship movement results in the elimination of the weakly bonded foulers by shear stress. Hence, employing hydrophobic FR surfaces of PDMS and fluorinated polymers with minimum surface tension presents potential use for highly-active ships and fast ferries. Numerous techniques were used to produce these potential surfaces since the first synthetic superhydrophobic surface was introduced in the mid-1990s [142]. Many researchers studied the chemistry of non-stick and anti-adhesion coatings based on Cassie-Wenzel hypothesis [143–145]. Recent advances in the development of water repellent nanocomposites for fouling prevention have been reported [138].

Superhydrophobic coatings prepared on various substrates have been reported using two different approaches: (i) application of suspension of suitable NPs in PDMS over a smooth surface, and (ii) infusing a lubricant liquid into a porous rough topology called slippery liquid-infused porous surfaces (SLIPS) [146]. These processes can result in the formation of a rough hierarchical binary microstructure and nanostructure with low surface energy and high water-repellent properties. Superhydrophobic PDMS-silica nanocomposite coating can become oil repellent (oleophobic) through surface modification by using fluoroalkylsilane (FAS) via a combination of their simple and scalable deposition processes, low surface energy, and roughness characteristics of aggregates [147]. Incorporating hydrophobic NPs into a polymer matrix is a practical method to artificially mimic lotus leaves. Extensive studies have been performed to fabricate superhydrophobic surfaces based on inorganic fibers, such as SiO_2 , ZnO , TiO_2 , and Fe_3O_4 -filled carbon nanofibers. This technique provides a versatile method to coat substrates of various sizes, geometries, and surface chemistries. Zhang et al. [148a] fabricated several ZnO superhydrophobic surfaces by utilizing oxidation reaction between Zn powder and H_2O to produce multi-scale structures, from microstructure to nanostructure. PDMS coatings incorporated with ZnO/SiO_2 NPs exhibit outstanding superhydrophobicity and lotus effect characteristics [148]. Superhydrophobic surfaces with a water CA of 158.3° are readily prepared by mixing silicone resin, aminopropyltriethoxysilane, and Fe_3O_4 NPs [149]. Wen et al. [150] prepared superhydrophobic coating by modifying PDMS with micro- CaCO_3 /nano- SiO_2 composite particles with hexamethyldisilazane and further blending the modified particles with mixed PDMS and *c*-aminopropyltriethoxysilane. Li et al. [150c] prepared superhydrophobic coatings from organosilane-modified nano SiO_2 particles. Xu et al. [151] combined perfluorooctanoic acid-modified TiO_2 NPs with polystyrene (PS), while Harton et al. blended PDMS-modified SiO_2 with PS to achieve the superhydrophobic coatings as reported in the literature [152]. Immobilizing the photocatalytic activity of TiO_2 by SiO_2 NPs is one strategy to prevent the release of photocatalysts from coatings. Other superhydrophobic coatings were prepared by directly modifying PDMS resin with unmodified magnetite (Fe_3O_4) NPs.

One-dimensional (1D) nanomaterials with high surface area and unique morphology are particularly appealing in the fabrication of highly efficient self-cleaning coating applications. Surfaces coated with 1D nanostructure, such as aligned nanotubes, nano-fibres, nanorods, and nanowires, have exhibited excellent superhydrophobicity, with a CA larger than 150° [153].

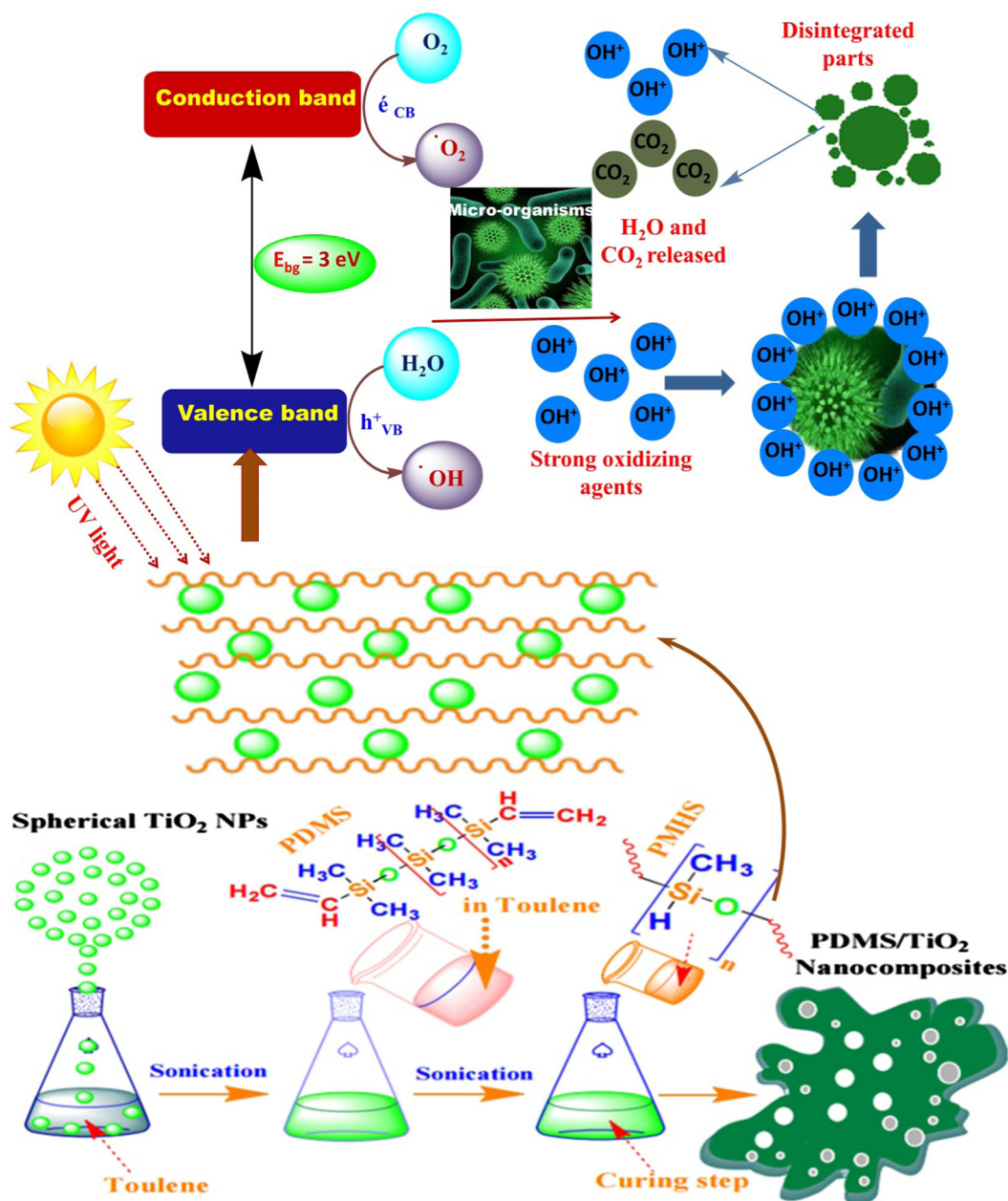
CNT forest functionalized by poly(tetrafluoroethylene) is also composed of a superhydrophobic surface. Aligned silicon carbide nanowires show excellent superhydrophobic property after they are functionalized with perfluoroalkylsilane. The superhydrophobicity characteristic of the composite film of CNTs and silicon carbon nanowires film could be tunable by controlling the ratio and different morphology [153,154]. Fluoro-compounds and resins have also been widely explored because of their low surface tension. Superhydrophobicity is induced by roughening the fluorinated surface with low-surface-energy materials [155].

Aizenberg et al. [156] demonstrated the performance of SLIPS. The slippery materials were obtained by infusing a viscous liquid (perfluorinated lubricants [157] or silicone oil [158]) into porous or textured substrates. The infused liquid was maintained as a thin dynamic layer at the substrate to form hydrophobic interface. The created interface allows other fluids to slip through the surface with a sliding angle of $<2^\circ$ [159]. Microporous or micro-structured surfaces, such as poly(tetrafluoroethylene) (PTFE) and fluoro-silanized surfaces, were used as SPLS substrates. SLIPS shows fouling resistance against various fouling communities [160]. Applying PTFE obtains higher AF performance than applying PEG-modified surfaces [18]. This finding is related to the high mobility on the slippery interfaces and low surface energy of the interfused liquid. However, SLIPS may be a promising candidate when using fluorinated lubricant PTFE in maritime ecosystem. Zhang et al. [161] reported a simple pathway to form superhydrophobic surfaces by extending a film of the waxy PTFE with the addition of fibrous crystals of a large fraction of void space. In summary, the blending technique is an essential avenue to create outstanding superhydrophobic surfaces because of the limited solubility of fluoropolymers.

7.2. Photo-induced superhydrophilic FR nanocomposite coatings

The utilization of UV and sun rays for reinforcing the AF performance on ship hulls has driven worldwide concern because of economic and ecological considerations [162]. Changing the surface wettability of a designed nanocomposite to a superhydrophilic character ($\theta \leq 10^\circ$) is the main function achieved by UV light irradiation. Photocatalytic activity of definite nanofillers could be utilized for decreasing fouling settlement and maintenance costs by reducing surface free energy for costly self-cleaning operations (Fig. 14) [162]. TiO_2 NPs have gained considerable attention because of their unique physico-chemical and mechanical characteristics, availability, and potential AF capability. Nano- TiO_2 particles, as semiconductive materials, semiconductive materials perhaps the most widely used among photocatalysts because of their stability, low cost, strong oxidizing agent, and promising UV light sensitivity nature [163]. When TiO_2 NPs are exposed by UV illumination from the solar light, two types of generated carriers (electrons and holes) are obtained (Fig. 14A). These carriers are produced by photon energy ($h\nu \geq$ the energy band gap (E_{bg}); their values are usually 3.0 eV for rutile or 3.2 eV for anatase at a wavelength of 388 nm. The photo-generated electrons (e^-) transfer from the valence band to the conduction band, while the holes (h^+) remain in the valence band. The e^- in the conduction band can reduce oxygen to superoxide anion radicals, while the strong oxidizing capability of h^+ can oxidize water to produce a hydroxyl radical. As a consequence, this radical can remove adsorbed organic pollutants and easily clean nanocomposite surfaces [162]. The utilization of dispersed TiO_2 NPs for designing self-cleaning and non-stick FR nanocomposite surfaces had been reported [129,162]. The non-toxic FR behavior of well-dispersed and spherical single crystal TiO_2 in PDMS nanocomposites presents evidence of the formation of non-toxic FR behavior. This design inspiration was used to improve coating smoothness, weak fouling attachments, surface stability against biodegradation, and superhydrophilic character induced by UV irradiation [129]. However, TiO_2 NPs can only absorb UV light because of their wide band gap that constitutes 4% of the sun rays [163,164]. Thus, enhancing the photocatalytic activity of TiO_2 nanocomposite coatings to cause absorption in the visible region is an attractive route for pollutant degradation and fouling elimination (Fig. 14 and Scheme 2). The application of noble-metal/ TiO_2 photocatalysts for superhydrophilic self-cleaning coating achieves important economic and ecological returns. Doping with selected noble metal, such as Au or Ag, is another strategy to enhance the photocatalytic activity of TiO_2 NPs [164]. For example, Au- TiO_2 NPs show a high photo-activity toward superhydrophilic self-cleaning coating designs [165,166]. Chen and Carroll [167] synthesized Ag@ TiO_2 , which showed an excellent photocatalytic performance by degrading Rhodamine B compared with the commercial TiO_2 . The photocatalytic performance was enhanced by completely degrading Indigo carmine with Ag-supported TiO_2 for 2 h under visible light irradiation [168]. Ag NPs are very chemically reactive and thus are oxidized upon direct contact with TiO_2 . Therefore, using SiO_2 or Al_2O_3 NPs as interlayer between TiO_2 and silver NPs may prevent the oxidation of Ag to modulate the distance between Ag and the TiO_2 NPs [165]. Ag- SiO_2 - TiO_2 photocatalysts have been developed to obviate silver NPs oxidation. The self-cleaning phenomenon of an Ag- SiO_2 - TiO_2 nanocomposite coating was proven to have superior photocatalytic properties [165]. Surfactants such as n-octylamine added to the mesoporous nanocomposites can improve the photocatalytic performance by coarsening the pore structure [166].

However, incorporating TiO_2 photoactive NPs in organic layers is promising for self-cleaning applications; a drawback of oxidizing the polymeric resin should be considered [166]. Consequently, changes in photo-induced wettability in TiO_2 NPs increased the difficulty of developing water repellent properties compared with water-loving superhydrophilic surfaces. Adsorption of the water-repellent fluoroalkylsilane polymer on TiO_2 nano-surface was studied and could enhance durability



Scheme 2. Synthesis and photo-induced superhydrophilic TiO₂/PDMS FR nanocomposite surface [125]. Copyright 2016, reproduced with permission from Elsevier.

and superhydrophilicity [166]. FR nanocomposites based on mixed TiO₂/noble metal photocatalyst with hydrophobic polymers have been rarely studied, and future studies on this state of art technology are thus promising.

7.3. Amphiphilic FR coatings

The complex biochemical phenomenon was confirmed by studying the mechanism of biofouling. For example, water repellent adhesive protein binds barnacles, whereas diatoms bind through hydrophilic proteins. Hydrophilic coatings minimize fouling attachments and adsorption of protein and thus provide potential options for utilization as FR coatings [169]. Combining the properties of hydrophilic and hydrophobic materials results in effective new FR terminology called amphiphilic surfaces (Fig. 17). These surfaces with prominent nanoscale hydrophilic and hydrophobic domains are new generation for non-biocidal coatings that possess surface properties which prevent the binding of biological macromolecules and

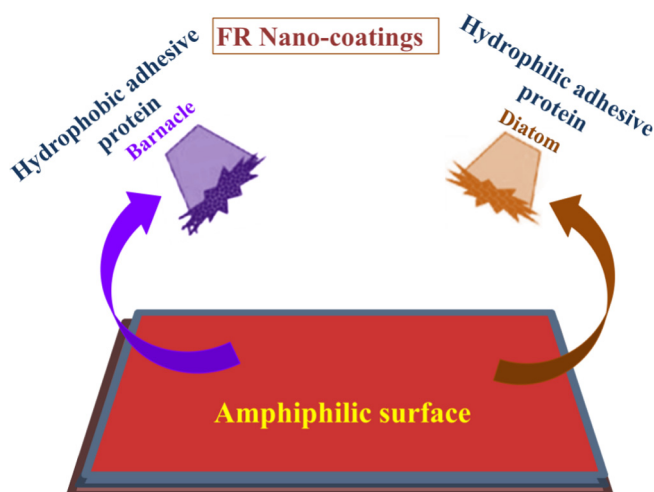


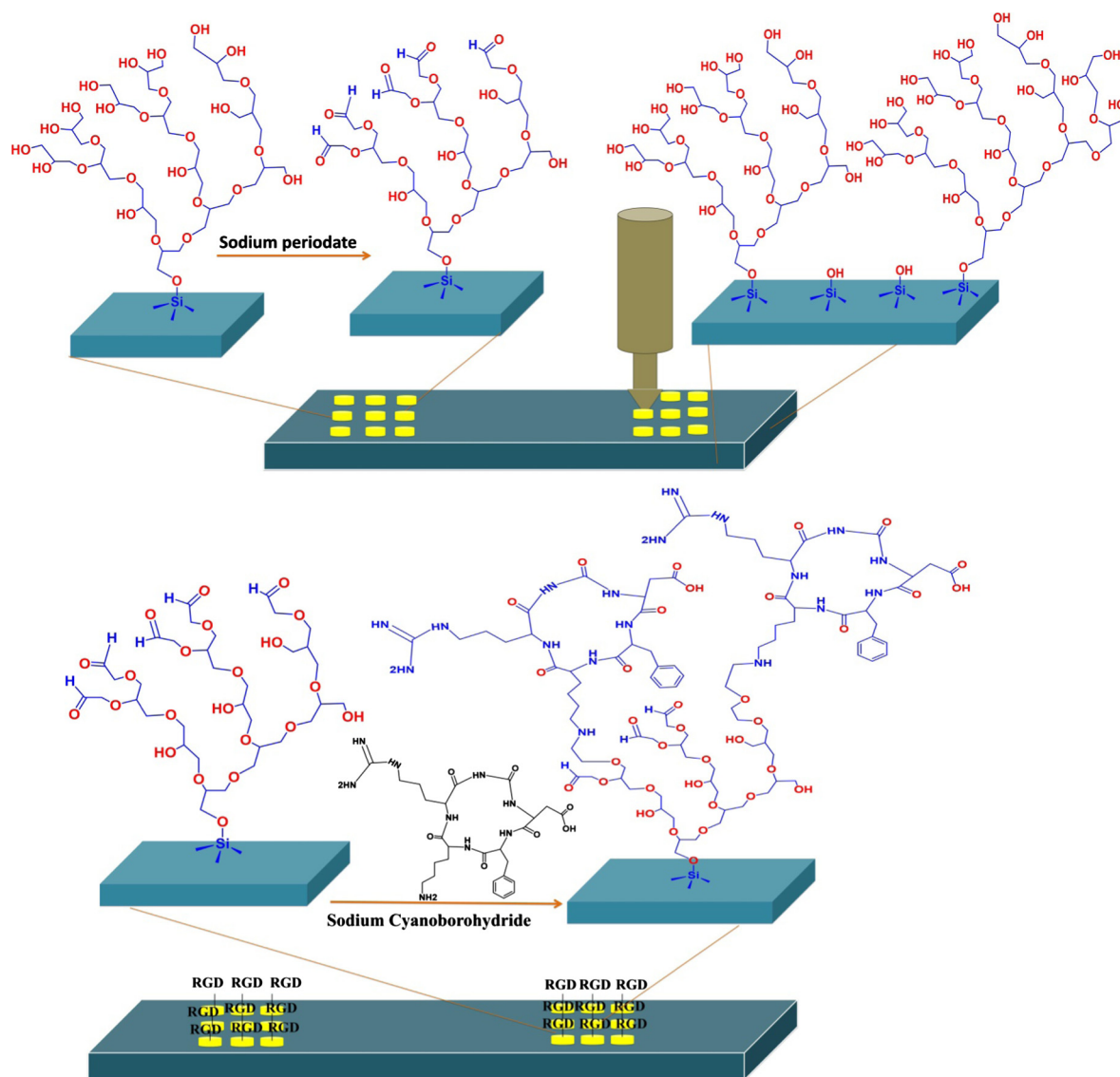
Fig. 17. Amphiphilic nanocomposite surface for fouling prevention (prevent both hydrophobic and hydrophilic adhesive protein) on the immersed objects.

improve fouling-prevention characteristics [170]. Fluoropolymer-modified silicone nanocomposites enjoy amphiphilic surfaces and block copolymer structure, providing both protein adsorption resistance and low surface energy characteristics [171]. A methoxy-terminated PEG conjugated with dihydroxyphenylalanine shows prominent amphiphilic FR performance over a virgin silicone coating [5,170]. Based on its water solubility, neutral nature, ingrained hydrophilicity in AF performance and flexible chains, PEG gained unparalleled capability to be hydrated with water. The coordination capability of PEG as well as steric exclusion effects are essential factors for fouling prevention and protein adsorption resistance [172]. PEG monolayers, multilayers and PEGylated hydrogel paints chains, have been evaluated in marine AF because of their bio-fouling anti-adhesion characteristics [173]. Hyperbranched polyglycerol (HPG) was also introduced and confirmed to have superior AF properties. HPG ensures an AF performance equivalent to PEG, but HPG provides more benefits, such as increased stability against oxidation and high functionality. Moreover, HPG activation and bioconjugation can be accomplished easily through a one-pot reaction; thus, the number of preparative steps and time consumption are reduced [174] (Scheme 3).

An effective FR paints were reported based on combination between hyperbranched amphiphilic fluoropolymers and PEG linear polymers and approved superior anti-adhesion of green *Ulva* sporelings compared with silicone paints [173]. Also amphiphilic block copolymer coatings based on grafted ethoxylated fluoroalkyl groups were studied and approved to release fouling organisms easily. The results suggested that the tailored amphiphilic coatings can prevent algae settlements [173,175]. Studies have been performed to develop a triblock amphiphilic copolymer coatings that combine resistance to cell adhesion and low surface energy by randomly incorporating discrete PEG contain fluorine groups with polystyrene-block-poly(ethylene) blockpolyisoprene. The xerogel coatings exhibit anti-adhesion and FR effects toward various microorganisms. Barnacle cyprids show reduced settlement on hydrophobic xerogel coatings than on hydrophilic coatings, but the removal percentage of diatoms *Nitzschia perminuta* increases as surface wettability is increased [175]. Phosphorylcholine (PC) copolymer films can be formed from hydrophilic PC block and hydrophobic lauryl block networks. A PC film used without pre-treatment promotes the adhesion of *Nitzschia closterium* because of the presence of hydrophobic lauryl groups with good AF/FR properties at the film/water interface. The frequently observed protein resistance of some nanopatterned, amphiphilic diblock copolymers is due to the intrinsic high density of surface interfacial boundaries [176].

7.4. Graphene-related materials in self-cleaning FR coatings

Modeling innovative and non-toxic FR nanomaterials for vessel bottoms is a challenge [177,178]. Carbon nanostructured materials have also been extensively investigated [179]. Graphene materials, including graphene, graphene oxide (GO), reduced graphene oxide (rGO), and graphene-embedded polymers are the most widely used nanomaterial in the field of biotechnology and environmental technique [180]. Graphene materials possess characteristics such as economical savings, biocompatibility anticorrosive performance, and environmental security, thus presenting potentials for various applications [181,182]. These carbon-based compounds are 2D nanomaterials with sp^2 -hybridized carbon atoms in a hexagonal lattice model and are featured with supreme mechanical properties and large surface area [179,183]. Great efforts have been paid to incorporate graphene nanomaterials in polymer matrix for developing AF coatings for the next generation [180,184–186]. Studies on the AF performance of Graphene, GO, and rGO have demonstrated superior properties of rGO to other materials because of the high surface area of the rGO nanomaterials [187,188]. Modern studies have investigated the anti-adhesion performance of nanomaterials based on graphene compounds which inhibit protein adsorption and biofilm formation [180,189].



Scheme 3. Amphoteric surface of HPG with diol terminated multifunctional groups that can be easily modified Sodium Periodate and thus controlled through further modifications and grafting [174a]. Copyright 2014, reproduced with permission from the American chemical society.

Nowadays, graphene-related materials are introduced as eminent catalyst supports in photocatalytic systems as they have the following characteristics [179,180]:

- Eligibility to accept electrons from photocatalyst and inhibit recombination of photo-generated holes.
- Extremely large surface area because of their unparalleled 2D structure.
- Narrow band gap energy, thus, any tailored graphene nanocomposites enjoy wide light absorption.
- High mechanical strength to stabilize the photocatalyst and also highly conductive.

Thus, modern studies have reported that anchoring well defined photo-catalysts on the surface of graphene compounds introduces nanocomposites with superior photocatalytic behavior for various applications [190,191]. Many graphene-based nanocomposite nanomaterials, like TiO_2 -GO, ZrO_2 -GO, ZnS -rGO, and CdS -graphene- TiO_2 have been demonstrated and obtained higher photocatalytic performance than the unanchored metal oxides [192–194]. Also, He [193] reported the development of inexpensive (ZnO -rGO) nanocomposite films with improved photo-induced and superhydrophilic characteristics for self-cleaning applications. This phenomenon is promising for solar light-boosted self-cleaning applications. Selim et al. [129] reported eco-friendly FR coatings based on utilization of UV-irradiation for boosting self-cleaning behavior of PDMS/spherical single crystal TiO_2 nanocomposites. The utilization of solar light boosting for smart FR coatings based on rGO

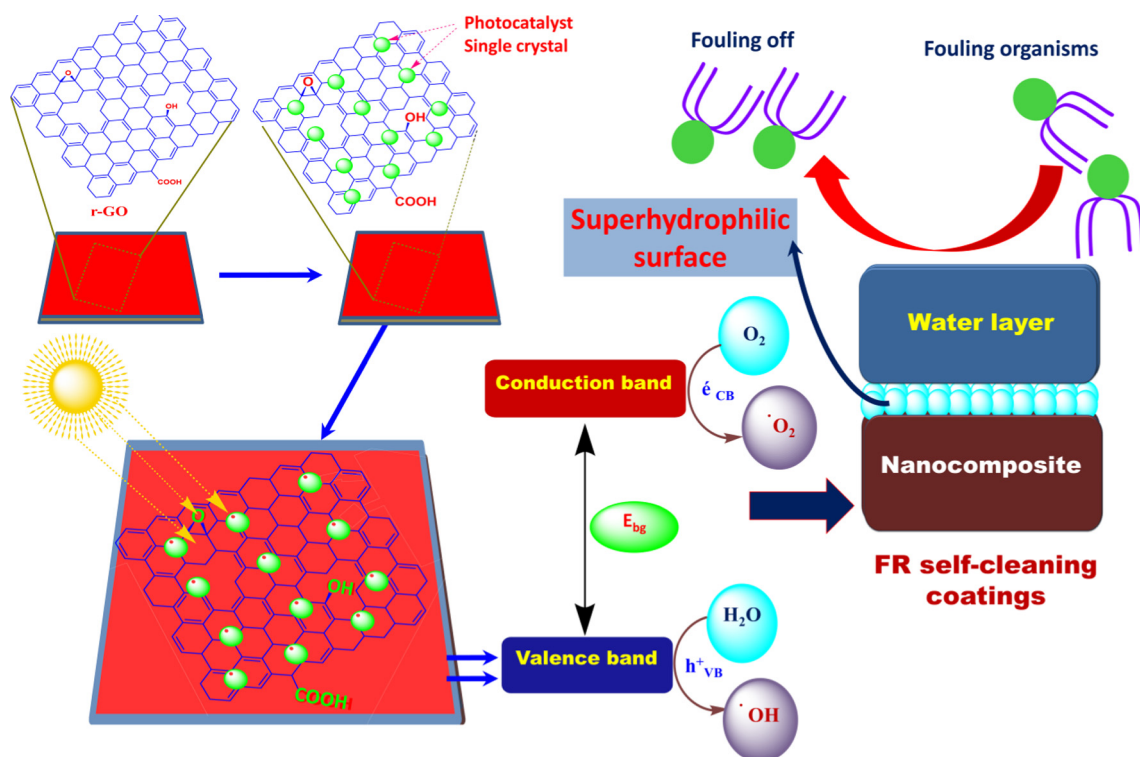


Fig. 18. Proposed FR mechanism by using solar light boosted self-cleaning graphene related materials (r-GO anchored with photocatalyst) coatings for applying in shipping industry.

anchored with single crystal photocatalyst materials is a future trend for superior FR and durability performance (Fig. 18) which has yet to be examined.

8. Outlook and future directions

Fouling represents a major disaster in the naval industry and exacerbates fuel drain, drag resistance, maintenance costs, and negative effects on the environment. The prevalent utilization of biocidal AF coatings has caused numerous environmental repercussions and resulted in strict legislations in the shipping industry. Non-stick silicone FR technology prevents the attachments of colonizing species and weakens their cohesion strength by providing low-friction and ultra-smooth surfaces. These surfaces are non-leachant and are fairly stable in water. FR paints based on PDMS elastomers exhibit excellent self-cleaning and AF performance; however, filler reinforcement has become necessary to reduce costs and overcome the mechanical disadvantages. Nano-engineered composites have a great potential for application as durable and eco-friendly alternatives to toxic AF paints in addition to minimizing fuel consumption and environmental pollution emissions. Consequently, many newly designed FR coatings are also based on organic-inorganic hybrid nanocomposite materials.

Superior criteria for marine FR coating composites are urgently needed, as follows:

- Economical savings, ecological considerations, and prolonged longevity on the basis of the costs spent annually for fouling
- Simplicity, non-toxicity, environmental sustainability, commercial feasibility, durability, and high performance FR surfaces
- Improvement in the maneuverability and increased time of intervals between dockings.
- Worldwide acceptance according to the IMO and international legislations
- Desirable self-cleaning surfaces with durability characteristics, low surface free energy, extremely smooth topology, low microroughness, and functionalized nature.

With the capability to tailor the FR performance, new series of hydrophobic, hydrophilic, and amphiphilic FR nanocomposites are designed as a modern stream for self-cleaning applications in maritime navigation. Techniques to controlling AF composition systems including fluorinated, silicone, PU, PCL, and PEG based FR coatings, in addition to the utilizations of hyperbranched and linear polymers and copolymers, show the greatest contributions to the improved FR properties. The

fundamental functions of size-and shape-controlled hybrid NPs can tailor the controlled hydrophobicity/hydrophilicity and surface free energy that produce ultra-smooth surfaces. To a high extent, GO-, rGO-, MWCNT-, and SWCNT-coated metal, metal oxide NPs, and multifunctional hybrid NPs are good models to provide an important guidance on the influence of hydrophobicity/hydrophilicity in designing high performance FR surfaces. In this filler design, the particle size, shape, and surface characteristic or well-dispersion of metal and metal oxide NPs in nanocomposite AF systems promote the physico-mechanical and wetting behavior. These two key factors effectively prevent agglomeration problems and improve the activity of silicone nanocomposite FR system. The application of a wide range of recent FR technologies in the marine industry is being considered to satisfy ecological and economic considerations. Some of these technologies have converted to commercial products, and others remain at the laboratory–development stage. Several surface factors, such as free energy, CA, elastic modulus, topology, and structure–property relationship, are essential to inhibit bonding of marine organisms.

This review discusses the future requirements of nano-FR innovations applicable to marine environment. A new class of nano-FR technology is continually progressing in for future ecological and economic applications. Synthetic nanomaterials with control over the structural morphology, geometry, tunable particle size and surface functionality, and desired sites can promote their capability as potential candidates for the production of eco-friendly AF nano-coatings. Organic-inorganic nanomaterial hybrids can also aid in the improvement and development of economically viable high-performance nano-FR composite coating. It is marine environmental security that underpins less of a potential hazard and thus a more attractive route is the development of innovative nano-FR materials, which possess a wide range of compositions, structural features, and surface properties such as:

- 1D nanostructured materials can provide promising and supreme superhydrophobic self-cleaning surfaces because of their extremely high surface area and unparalleled morphology. Thus, the incorporation of 1D nanomaterial in non-toxic FR coatings is a strategic goal for fouling prevention.
- Multifunctional nanohybrid fillers, such as CNT/Ag/ZnO and Ag/Al₂O₃/TiO₂, have superior photodegradation properties, which are essential for solar light-boosted self-cleaning surfaces. This nano-FR design enables the utilization of solar light, as a clean and renewable source of energy, to promote non-toxic FR coatings.
- Graphene hybrid photo-catalyst associated with highly exposed crystal facets makes an excellent candidate for the manufacture of sun-induced photo-nano-FR coatings. The in situ synthesis of photo-FR nanocomposite filler enhances physico-mechanical and FR properties. This design is a projected trend for superior AF and durability performance.
- Triblock copolymer/fluorine-free amphiphilic FR coatings with high resistance to fouling adhesion as well as the desirable surface and durability characteristics would act as attractive materials for the coming generation of antifouling system.
- The synergistic effects of the hydrophilic-ended groups of hyperbranched polymer structures, and physical/chemical/mechanical, and morphological surface features are vital for sustained FR devices.

These characteristics of NPs are expected to save billions of dollars annually for the treatment of fouling problems. With the diversity of these organic and inorganic hybrid nanocomposites, innovative trends may be promising in minimizing fuel consumption, maintenance costs, and docking intervals. The FR models summarized in this comprehensive review are relevant examples of how materials technology can help solve global environmental concerns caused by outdated or previously applied chemical technologies. The results of this study provide new insights into the manufacturing of scalable, cost-effective, and reliable FR nano-based formulation coatings. Furthermore, the utilization of renewable solar light in engineering FR coatings and in protecting aquatic environments can be feasible.

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